

With More Than Shovels

A Study on the Risk-
Benefit Paradox at the
Center of Nuclear Civil
Defense

MADELINE BERZAK,
NOAH PETERS,
NOAH BURDICK AND
ADA MELTZER



About XLab.

The Existential Risk Lab (XLab) is a University of Chicago research institution working to reduce the risk of human extinction, with a focus on the pressing threats of superintelligent AI and nuclear war.

XLab studies the risks that threaten to end humanity, communicates them to a broader audience, and trains the next generation of experts to address them. Although the field receives relatively little funding or academic attention, the near-term possibility of human extinction is widely recognized by experts. Leading AI researchers, Turing Award winners and leaders of frontier AI labs have publicly characterized AI as an extinction-level risk (Center for AI Safety, 2023). In XLab's own assembly on nuclear risk in 2025, over 120 Nobel Laureates signed a declaration recognizing the need for immediate action to prevent the threat of nuclear war.

The existential risk research community is small, pre-paradigmatic and underfunded compared to more established research communities. XLab's goal is to accelerate this field into a robust science. We focus on ambitious meta-science and conceptual research that forms the foundation of how we study large-scale catastrophic risks. Our training programs aim to cultivate good epistemics and scientific skills in students by advising and supporting them through scoping and executing their own research projects. Finally, we communicate and collaborate with the scientific community and aim to produce research that directly informs policy decisions.

About XLab Working Groups.

XLab's research working group program allows students to pursue impactful research over quarter-long project cycles advised by XLab staff. We provide otherwise rare opportunities for students to engage with topics in nuclear risk, familiarize themselves with the research process, and develop critical analysis skills over the course of the project. Each working group produces a journal article-length deliverable.



With More Than Shovels: The Risk-Benefit Paradox of a Modern Nuclear Civil Defense

Madeline Berzak* Noah Peters† Noah Burdick‡ Ada Meltzer§

XLab Research Working Groups
Winter 2026

Abstract

Nuclear risk has arguably intensified as the international security environment enters a tripolar era marked by deteriorating arms control regimes, force modernization, and compressed decision-making timelines. Yet civil defense, once a pillar of U.S. nuclear preparedness, has largely faded from policy agendas and the academic literature on risk since the late Cold War. Interest from effective altruism-aligned foundations on “right-of-boom” interventions research has reopened the conversation of whether civil defense merits reinvestment, although the nuclear policy field lacks a modern analytical framework for evaluating it.

This research argues that nuclear civil defense poses an understudied paradox: interventions designed to reduce expected casualties may simultaneously alter the probability of nuclear war itself in a way that the net expected value depends on the relative magnitude of these countervailing effects. This paper develops a theoretical framework that the field lacks, integrating robust decision-making under deep uncertainty and complexity, with cost-benefit analysis of specific classifications for interventions.

This paper considers that traditional Cold War civil defense measures (e.g., public shelter construction, mass urban evacuations, and educational campaigns) likely carry negative expected value under modern risk conditions, while a reconceptualized intervention portfolio centered on agricultural resilience, pharmaceutical and dosimeter stockpiling, trade-relationship hardening, and post-detonation response capacity offer substantial positive returns at comparatively modest cost. More broadly, this paper considers the implications for the treatment of catastrophic risks under deep uncertainty and complexity and proposes tiered policy recommendations.

1 Introduction

Early civil defense logic was famously summarized by Reagan administration strategist T.K. Jones in 1982: “If there are enough shovels to go around, everybody’s going to make it.”¹ Through insisting that, “it’s the dirt that does it,” Jones cultivated a vision of survival that California Senator Alan Cranston eventually described as “far beyond the bounds of reasonable,

*mberzak@uchicago.edu

†npeters@uchicago.edu

‡noahburdick@uchicago.edu

§ameltzer@uchicago.edu

¹Robert Scheer, “Thinking Tuna Fish, Talking Death: Essays on the Pornography of Power.” *With Enough Shovels*. 1988.

rational, responsible thinking.”² Civil defense was grounded in the belief that nuclear war may not be as catastrophic as the field once feared. Despite the relatively absurd and bizarre nature of nuclear civil defense in the United States, the broader apparatus rarely reinterrogates the initiative beyond its strange impact on American culture—civil defense brought the risk of nuclear war to the forefront of public consciousness for the majority of the Cold War era. Yet nuclear civil defense during the Cold War was, at the time, an unprecedented national undertaking of massive proportions.³ Civil defense, a class of risk-reduction measures to increase nuclear preparedness in the civilian population, grew to encompass interventions including: the Eisenhower interstate system, Bert the Turtle’s “duck and cover” drills, educational initiatives, public fallout shelter systems, supply and food stockpiling, as well as periodic tests of the nationwide emergency broadcast system. The initiative’s effects stretched far beyond preparedness, touching on the psychological, strategic, economic, political, philosophical, and decision-making domains. Civil defense made the risk of nuclear war feel both imminent and proximate to the American public—then, it all but disappeared from American life, both as a policy domain and an object of public attention.

During the Cold War, civil defense constituted what anthropologist Joseph Masco has called an extraordinary national conversation about collective death; an extraordinarily powerful means of defining the boundaries of both security and threat for the public, while training the public to think about nuclear war in specific ways.⁴ This kind of cultural education and training often has material consequences. Civil defense did not merely present the possibility of survival in the event of a nuclear war, rather it actively shaped public perception of how likely, how survivable, and how “thinkable” a nuclear war was, when most believed it was safer for nuclear war to remain decidedly “unthinkable.” Nuclear strategist Herman Kahn argued that the refusal to think through post-detonation scenarios was itself the immoral position, because it left the public defenseless against failure modes of deterrence that were entirely within the realm of possibility.⁵ Kahn’s critics, including Thomas Schelling, argued that presenting the idea of nuclear war being survivable made it thinkable as a result and making it thinkable undermined the balance of deterrence itself, as it relied upon both sides fearing total destruction.⁶

Although its effects on signaling were largely ambiguous, the omnipresence of civil defense during the Cold War played a significant role in constructing the strategic environment in which deterrence operated. By making nuclear war more thinkable, civil defense also opened a still-unresolved conflation between extinction and civilizational collapse and exposed the public’s enduring preference for simple apocalypses over the uneven, complex, and asymmetric catastrophes a nuclear exchange would actually produce. This still unresolved strategic debate points to something more nuanced: civil defense poses a fundamentally understudied paradox: interventions designed to reduce expected casualties may simultaneously alter the probability of nuclear war, such that the net expected value depends on the relative magnitude of these countervailing effects. This paper develops a theoretical framework that the nuclear policy field lacks, integrating robust decision-making under deep uncertainty and complexity with cost-benefit analysis of specific classifications of interventions.

This research argues that traditional Cold War civil defense measures (e.g., public shelter

²Ibid.

³Joseph Masco, “The Future of Fallout and Other Episodes in Radioactive World Making,” Duke University Press, 2021.

⁴Ibid.

⁵Herman Kahn, “Thinking About the Unthinkable in the 1980s,” 1982, pp. 55.

⁶Thomas C. Schelling, *Arms and Influence*, (New Haven, CT: Yale University Press, 1966), pp. 241–242.

construction, mass urban evacuations, and educational campaigns) likely carry negative expected value under modern nuclear risk conditions—compressed post-launch decision timelines, precision weapons, and advanced delivery systems—while a reconceptualized portfolio centered on agricultural resilience, pharmaceutical and dosimeter stockpiling, trade-relationship hardening, and post-detonation response capacity offers substantial positive returns at comparatively modest cost. This paper also develops tiered policy recommendations and a discussion of implications for the broader analysis of catastrophic risk interventions under deep uncertainty and complexity.

The following background traces the rise and fall of U.S. civil defense while providing background on the moral hazard critique of civil defense and the contemporary nuclear risk landscape. Section 3 formally develops the risk-benefit paradox, identifying the mechanisms through which civil defense affects the probability of war and the conditional factors governing their magnitude. Section 4 discusses the methodologies used in this research, pairing robust decision-making under deep uncertainty with cost-benefit analysis. Section 5 demonstrates why traditional civil defense measures carry negative expected value under modern conditions. Section 6 introduces the reconceptualized intervention portfolio and its cost-benefit analysis. Sections 7 and 8 present findings, tiered recommendations, and a discussion of nuanced implications, including our response to the “moral hazard” critique and how this research is positioned within the broader right-of-boom debate.

2 Background: A Question Unresolved

Conceptually, civil defense itself is both a policy domain, and a cultural phenomenon; a complex network of interventions that elicit a wide range of reactions. In spending over a year documenting the history of civil defense, U.S. historical consultant Harry Yoshpe noted that the experiences he was recounting were not happy ones, as “the problems of planning for survival and recovery from nuclear attack defy full comprehension and easy solutions.”⁷ Civil defense initiatives in the United States devolved from being perceived as a vitally important part of U.S. Cold War nuclear doctrine, to an absurdly bizarre cultural relic of a bygone era.

2.1 The Rise and Fall of U.S. Civil Defense

The rise of civil defense began conceptually during World War II. The blitz on Britain, air raids in Germany, and firebombing of Tokyo demonstrated the potential of countermeasures, and led President Roosevelt to create The Office of Civilian Defense (OCD) in 1941 to protect against potential enemy attacks by mobilizing volunteers and coordinating different levels of action. The Cold War expedited the development of civil defense at a broader scale, in part because of the sudden onset of the real possibility of fighting a nuclear war. The Federal Civil Defense Act of 1950 marked the beginning of this re-expansion, amidst rising tensions and uncertainties with the Soviet Union testing their first nuclear weapon in August of 1949, and China entering into the Korean war. The act intended to create and bolster urgency for national preparedness within the U.S. population, and established frameworks for operation that heavily relied on participation, cooperation, and adoption by the general public: the federal government

⁷Harry B. Yoshpe, “Our Missing Shield: The U.S. Civil Defense Program in Historical Perspective.” Report Contract No. DCPA 01-79-C-0294. Federal Emergency Management Agency (FEMA). April 1981. apps.dtic.mil/sti/tr/pdf/ADA099634.pdf

would provide “necessary coordination and guidance,” and manage the Federal Civil Defense Administration, while the bulk of responsibility for carrying out policies was vested in “states and their respective political subdivisions.”⁸

Beyond the geopolitical tensions that drove direct investment, civil defense was heavily dependent on political will sustained at every level—among officials, decision-makers, and the general public. Much of this political will was actively manufactured. The broad consensus among experts at the time held that even a full-scale war between the USSR and the United States would leave 60 to 80 million American survivors.⁹ An official government pamphlet on Survival Under Atomic Attack asserted as much: “You can live through an atom bomb raid... The secrets of survival are: know the bomb’s true dangers. Know the steps you can take to escape them.”¹⁰ Effective civil defense relied heavily on volunteer participation from the public and on state governments, thus, the risk had to be communicated as both tangible and immediate, while the government simultaneously communicated that nuclear war was survivable.¹¹ When the dominant narrative shifted from survivable attack to global and irreversible collapse, civil defense no longer made logical sense as a public investment. Once the program lost its bureaucratic home, executive support, and coordination, political will eroded—gradually, and then all at once.

Towards the end of the Cold War and in the post-war period, civil defense was largely abandoned, and most U.S. civil defense programs concluded after three decades of operation. The shift immediately followed the collapse of the Soviet Union, but began when the Federal Emergency Management Agency (FEMA) was established in 1979 during the Carter administration to oversee “emergency management and civil defense.”¹² FEMA illustrated a shift away from focusing on nuclear preparedness to taking an “all-hazards,” collective security approach to managing natural and manmade disasters and localized emergencies rather than nationwide radioactive fallout.¹³ The official end of civil defense occurred in 1994 during the Clinton administration, when Public Law 93-337 repealed the Federal Civil Defense Act of 1950.¹⁴ This process was part of the gradual restructuring of post-Cold War national security priorities. When the act was repealed, it transferred and reorganized any remaining responsibilities to other administrative bodies, including FEMA. Beyond organizational and budgetary constraints, the advent of higher-yield thermonuclear weapons meant that even if a warning system detected an incoming attack, the destruction would be too immediate and overwhelming for traditional civil defense to be as effective as proponents once thought. Amidst almost constant debate over utility and strategic weight as well as decreasing public buy-in with increasing skepticism on nuclear risk, civil defense was seemingly always going to come to an end.

⁸“United States Statutes At Large: Containing the Laws and Concurrent Resolutions Enacted During the Second Session of the Eighty-First Congress.” The Federal Civil Defense Act of 1950. *United States Congress*. Vol 64. 1950-1951. [hsdl.org/c/abstract/?docid=456688](https://www.hsdl.org/c/abstract/?docid=456688)

⁹Richard E. Sincere, “Civil Defense: A Moral, Political, and Strategic Approach.” *The Journal of Social, Political, and Economic Studies*. Washington D.C. Winter 1984.

¹⁰“Survival Under Atomic Attack: The Official U.S. Government Booklet.” Executive Office of the President, Civil Defense Office. 1950. (accessed through Oak Ridge National Laboratory).

¹¹*Ibid.*

¹²Sarah Ahmad and David Wessel, “What does the Federal Emergency Management Agency (FEMA) do, and how is it funded?” *Brookings Institute*, June 2025. [brookings.edu/articles/what-does-fema-do/](https://www.brookings.edu/articles/what-does-fema-do/)

¹³“Chapter 1: The History of FEMA,” The Federal Emergency Management Agency Publication 1, The Federal Emergency Management Agency, 2010. biotech.law.lsu.edu/blog/pub.1_final.pdf

¹⁴Public Law 103-337 Repeal of Civil Defense, *U.S. House of Representatives*, 1994.

2.2 The Moral Hazard Critique

Cold War era strategic debates remain largely unsolved.¹⁵ Those debates were principally based on strategic grounds: whether population protection was technically futile, or whether it was destabilizing. Woven in these debates was a more durable objection; one that is moral rather than strategic in character and has come to be critiqued through the language of moral hazard. While the critique is old, the label is new.

In its original actuarial definition, moral hazard names the tendency of insurance against a loss to make the insured behave in ways that raise the likelihood of incurring it; the migration of the concept from underwriting into normative argument is itself contested.¹⁶ On nuclear risk, the definition translates smoothly: the moral hazard critique argues that advocating for and investing in post-detonation interventions creates both a moral and structural hazard that in essence, rewards failure.¹⁷ Establishing effective and robust response mechanisms to manage the aftermath of a nuclear detonation (i.e., advanced fallout mitigation, massive humanitarian relief networks, and economic recovery blueprints, etc.) unintentionally lowers the perceived cost of nuclear conflict, making the use of nuclear weapons more likely. In other words, an insurance policy for the unthinkable.

The moral hazard critique has become commonplace within the nuclear risk apparatus, and its existence is perceived as legitimate even by those advancing the case for post-detonation work.¹⁸ It is important to note that the critique has rarely been expressed in a form that is precise enough to test, and where analogous claims have been examined empirically (in extended deterrence and domestic hazard mitigation) the findings are mixed rather than validating.¹⁹ Section 3 conceptualizes this mechanism formally as the risk-benefit paradox; the discussion and analysis of the paradox and a reconceptualized civil defense offer a response.

This research exists as a counterpoint to the current discourse on right-of-boom neglectedness within the broader nuclear risk apparatus.²⁰ While one can argue that the problems with right-of-boom research are structural, in many ways they are socio-historical and cultural: we do not consider post-detonation because it was abandoned by nuclear strategists during the Cold War. As a result, society also assigns an extremely narrow definition to “civil defense” as an absurd cultural past-time of the Cold War era instead of thinking about it as a broader concept or class

¹⁵The Cold War strategic debate is discussed in more detail in section 3, and is not rehashed here. In summary, the central division was between critics who held that population protection was technically futile, and critics who argued that it was destabilizing. They believed that effective damage limitation (civil defense) would erode mutual vulnerability, signal intent for a first-strike, and invite preemption based on survivability. See William H. Kincaid, “Repeating History: The Civil Defense Debate Renewed,” *International Security* 2, no. 3 (Winter 1978): 99–12; Jennifer Leaning and Langley Keyes, eds., *The Counterfeit Ark: Crisis Relocation for Nuclear War* (Cambridge, MA: Ballinger, 1984); Herman Kahn, Erich Fromm and Michael Maccoby, “The Question of Civil Defense — A Debate,” *Commentary*, 1962.

¹⁶Tom Baker, “On the Genealogy of Moral Hazard,” *Texas Law Review* 75, no. 2 (1996): 237–292. repository.law.upenn.edu/Documents/Detail/on-the-genealogy-of-moral-hazard/401175; Brett V. Benson, *Constructing International Security: Alliances, Deterrence, and Moral Hazard* (New York: Cambridge University Press, 2012).

¹⁷Tom Baker, 1996; Arnold Dembe and Leslie Boden, “Moral Hazard: A Question of Morality?” *New Solutions* 10, no. 3 (2000): 257–279. pubmed.ncbi.nlm.nih.gov/17208855/

¹⁸Christian Ruhl, “Philanthropy to the Right of Boom,” *Founders Pledge*, February 2023, founderspledge.com/research/philanthropy-to-the-right-of-boom.

¹⁹Benjamin E. Schwartz, *Right of Boom: The Aftermath of Nuclear Terrorism* (New York: Overlook Press, 2015); Ruhl, “Philanthropy to the Right of Boom,” 2024.

²⁰The relative neglect of right-of-boom research and interventions in nuclear risk primarily exists in Effective Altruism-aligned circles and was first popularized by researcher Christian Ruhl in early 2023.

of risk interventions.

2.3 The Contemporary Risk Landscape

The strategic environment into which any renewed civil defense would arrive differs fundamentally from the bipolar world that produced the original program. The international security landscape has entered a tripolar nuclear era: China’s arsenal has grown from roughly 250 warheads in 2015 to approximately 600 today, with the U.S. Department of Defense projecting 1,000 by 2030, ending the bipolar structure that organized Cold War deterrence—and with it, the analytical simplifications that structure permitted.²¹ Meanwhile, the bilateral arms control architecture that stabilized the late Cold War has not merely deteriorated but concluded. On February 5, 2026, the New Strategic Arms Reduction Treaty (New START) expired with no plan for replacement, leaving the United States and Russia without legally binding limits on strategic nuclear weapons for the first time since 1972.²² Russia’s offer to continue observing the treaty’s central limits informally went unanswered, and no successor negotiations are underway.²³ This points to structural significance: the prevention-focused architecture (e.g., multi-lateral treaties, verification regimes, sustained diplomatic channels, and arms control) that the field has treated as the primary vehicle for nuclear risk reduction can no longer be assumed as a stable backdrop in an emerging tripolar landscape. The burden that architecture carried does not disappear with it, rather it is simply reassigned and unexamined, to whatever remains.

The technical characteristics of a potential exchange have also evolved in parallel. Global arsenals are far smaller than their Cold War peak, sitting at ~12,321 warheads compared to a Cold War high of ~70,300 in 1986.²⁴ Today’s global arsenal remains civilization-threatening in scale, although the weapons themselves have advanced and modernized in ways that matter more for civil defense than raw numbers do. Precision delivery systems, hypersonic weapons, and modernized command-and-control have compressed post-launch decision timelines well below the three to six hours on which Kahn’s evacuation calculus depended.²⁵

In contrast, public engagement has not increased alongside intensifying nuclear risk. Attention to, and engagement with nuclear risk sits at historic lows compared to the Cold War era, consistent with the aforementioned cyclical pattern where attentiveness rises during acute crises and dwindles in peacetime calm. Despite the structural deterioration of the present risk landscape, it has produced no galvanizing crisis of the kind that historically translates abstract risk into tangible and proximate political will (e.g., the Nuclear Freeze movement in the 1980s).²⁶ The implementation trap referenced in the psychological literature, as a result, operates at full strength today.

²¹“Annual Report to Congress: Military and Security Developments Involving the People’s Republic of China.” *U.S. Department of Defense*. 2025. media.defense.gov/2025/Dec/23/2003849070/-1/-1/1/ANNUAL-REPORT-TO-CONGRESS-MILITARY-AND-SECURITY-DEVELOPMENTS-INVOLVING-THE-PEOPLES-REPUBLIC-OF-CHINA-2025.PDF

²²Matt Korda, et al., “The Aftermath: The Expiration of New START and What It Means For Us All.” The Federation of American Scientists (Washington, D.C.). February 2026. fas.org/publication/the-expiration-of-new-start/

²³Ibid.

²⁴Hans Kristensen, et al. “Status of World Nuclear Forces.” The Federation of American Scientists (Washington, D.C.). March 2025. fas.org/initiative/status-world-nuclear-forces/

²⁵Kelley M. Sayler, *Hypersonic Weapons: Background and Issues for Congress*, CRS Report R45811 (Washington, DC: Congressional Research Service, updated August 27, 2025), congress.gov/crs-product/R45811.

²⁶Lawrence S. Wittner, “The Nuclear Freeze and Its Impact,” *Arms Control Today* 40, no. 10 (December 2010), armscontrol.org/act/2010-12/nuclear-freeze-and-its-impact.

Reflecting the post-Cold War shift to an all-hazards approach, there is currently little funding or infrastructure allocated to civil defense as it was conceptualized during the Cold War. There are no plans for new public fallout shelters; investment has instead flowed toward strategic, technological, and infrastructure resilience, and certain government continuity facilities like Raven Rock and Cheyenne Mountain complexes remain in operation.²⁷ The demand that persists for Cold War-style civil defense is now largely consumer-driven in the U.S. private fallout shelter and bunker market that is projected to reach \$175 million by 2030, even as government disaster experts maintain that “bunkers aren’t necessary” and current FEMA guidance emphasizes sheltering in existing structures.²⁸ The demand for civil defense, in other words, did not disappear with the federal program, rather privatized: unequal, uncoordinated, and epistemically unmoored where the Cold War program was public, universal, and centrally organized. It is against this structurally deteriorated, technically transformed, publicly disengaged, and privately improvised landscape that the interventions examined in this research must be evaluated.

2.4 The Emerging Gap

While the historical literature on civil defense is rich, the nuclear field today lacks a robust framework for evaluating civil defense measures that takes seriously both its material effects on casualties and its constitutive effects on the probability of a nuclear war. The Cold War strategic debate over civil defense raised this question but could not conclude exactly how civil defense affects deterrence (i.e., stability vs. nuclear doctrine, whether it signaled prudent insurance or preparation for a first strike)—each of the aforementioned literatures grapples with this question, but steps around it. The debate ended in defunding, rather than resolution. There is a lack of scholarship that addresses the internal dynamics and conditions that led to the official end of civil defense in 1994, although the reasoning can mostly be inferred through ongoing debate on civil defense throughout the Cold War, as well as legislative and geopolitical changes.

Literature in the psychological domain found that protective infrastructure changes how populations perceive risk and respond to warnings, for better and for worse. The contemporary landscape shows preparedness demand migrating into private bunkers precisely because no public framework exists to absorb it. And the resilience literature, approaching from existential risk rather than strategy, independently rediscovered the old concern that visible preparedness can be read as hostile intent. Each literature registers, in its own vocabulary, the same structural feature: that civil defense does not merely respond to the probability of nuclear war but also influences it. However, no literature provides a framework for weighing this effect against the lives preparedness might save. The following section formalizes this structural observation as the risk-benefit paradox: the condition under which an intervention’s expected value depends simultaneously on the damage it mitigates and the probability it alters.

²⁷Terry Gross, “In the Event of Attack, Here’s How the Government Plans ‘To Save Itself,’” *Fresh Air*, NPR, June 21, 2017, [npr.org/2017/06/21/533711528/in-the-event-of-attack-heres-how-the-government-plans-to-save-itself](https://www.npr.org/2017/06/21/533711528/in-the-event-of-attack-heres-how-the-government-plans-to-save-itself).

²⁸Martha Mendoza, “Nuclear Bunker Sales Increase, despite Expert Warnings They Aren’t Going to Provide Protection,” Associated Press, December 2024, apnews.com/article/nuclear-bunkers-war-atomic-bombs-0356fa5b34067c138c64b9143f73c308.

3 The Risk-Benefit Paradox

While civil defense is most often designated to emergency management, it belongs principally to the strategic domain. Damage limitation, or trying to reduce the amount of harm an adversary can cause, has three factors: counterforce targeting, active defense, and the protection of population centers, which in other words, is merely civil defense. Snyder defines this as a denial capability operating within a deterrence system that relies on punishment and that is the source of its difficulty. What moves the probability of war, then, is not whether an intervention saves lives, but how the intervention is perceived by adversaries. They could view the signal as responsible insurance, or as preparation before launching a first-strike. Fundamentally, deterrence is about understanding perceptions. The Cold War debate over civil defense was merely defunded, and as a result unresolved. This section reconceptualizes it as a risk-benefit paradox in order to produce actionable analysis on the utility of civil defense.

3.1 Shortcomings of the Naive Calculation

The case for civil defense appears, at first, to be a matter of arithmetic. An intervention costs some amount, it saves some infrastructure and some number of lives in the event of a nuclear war, and the two figures are compared. The calculation appears simple because it holds one variable fixed. It assumes that the probability of nuclear war is a feature of the strategic environment, set by arsenals, doctrines, and diplomacy, and that preparedness merely determines how bad the war would be if it arrived. This is the naive assumption, and the previously reviewed history gives ample reason to reject it. That is, an intervention designed to reduce expected casualties may simultaneously alter the probability of the war itself, such that its net expected value depends on the relative magnitude of these countervailing effects. We call this the risk-benefit paradox, and this section develops it as a formal, analytical framework.

The failure of the naive assumption can be stated with more precision than the strategic literature has generally attempted. Consider a state willing to accept some constant expected damage $\bar{d}$ from a war it chooses to enter. Expected damage is the product of probability and damage, so this tolerance pins down the probability of war as a function of the damage that war would inflict:

$$\bar{d} = PD \implies P = \frac{\bar{d}}{D}. \quad (1)$$

The content of (1) is worth spelling out, because the remainder of the section is a study of when and how it fails. Suppose civil defense scales damage down by a factor $\lambda \in (0, 1)$, so that the war the state contemplates would now inflict λD rather than D . Under (1) the probability of war adjusts to

$$P' = \frac{\bar{d}}{\lambda D} = \frac{P}{\lambda}, \quad (2)$$

and the new expected damage is

$$P' \cdot \lambda D = \frac{\bar{d}}{\lambda D} \cdot \lambda D = \bar{d}. \quad (3)$$

Every unit of damage removed is repaid, exactly, in additional willingness to fight: halve the damage and the probability of war doubles, leaving expected damage untouched. Hence, (1) asserts that the probability of war is *unit-elastic* in damage,

$$\varepsilon \equiv -\frac{d \ln P}{d \ln D} = 1, \quad (4)$$

and unit elasticity is precisely the condition under which damage reduction accomplishes nothing: the expected damage $\bar{D}(D) = D P(D)$ satisfies

$$\frac{d \ln \bar{D}}{d \ln D} = 1 - \varepsilon = 0. \quad (5)$$

Equation (5) is the knife edge on which the whole question balances. If the probability of war falls off more slowly than one-over-damage ($\varepsilon < 1$), then expected damage rises with the size of the war a state courts, and reducing damage through civil defense reduces expected damage: preparedness pays. If probability falls off faster than one-over-damage ($\varepsilon > 1$), expected damage *falls* as wars get bigger (and therefore less likely), and reducing damage raises expected damage such that preparedness yields a worse expected outcome. The naive assumption (1) sits exactly on the edge.

The rationale is that a state pursuing a goal through deliberate entry into conflict has a maximum threshold for its expected losses, and behaves, in this respect, as a rational Bayesian actor.²⁹ If civil defense is treated in its most basic capacity, as a modifier or input variable to D , the consequence is immediate and unwelcome: civil defense is worthless in terms of reducing long-term expected damage. Whatever damage it removes, the state's tolerance restores through a proportionally higher willingness to fight. This is the paradox in its starkest form, and it is the null hypothesis the remainder of this section must contend with.

The paradox has a clear prior statement in the strategic literature. Ravenal's paradox of destruction holds that nations seeking to avert their own destruction in a possible war are incentivized to invest in damage limitation, through programs such as anti-ballistic missile systems or civil defense, yet doing so reduces the deterrent force bearing on them, since they can act with less fear of unbearable loss.³⁰ Reducing the cost of war increases the likelihood of war; reducing the likelihood of war increases the cost of war. What the Cold War literature never produced, as Section 2 established, is a framework for determining when this trade binds and when it can be escaped. Thus, the question this section pursues is whether there are modifications or permutations to the probability of nuclear war, related to civil defense, that alter the above relationship. This research identifies four and finds that the paradox binds only through some of them.

3.2 The Probability Shift

Treating the probability of nuclear war as insensitive to preparedness requires confidence that deterrence, once credibly established, is stable against changes in vulnerability. The record does not support such confidence. Tannenwald offers several reasons to doubt that mutually assured destruction (MAD) alone explains nuclear non-use. First, the doctrine cannot account for American restraint in the early Cold War, when the United States held preponderant capability and faced no assured retaliation. Second, the criteria for what constitutes sufficiently disincentivizing destruction are left unexplained, and the language of war being rendered unbearable was invoked, without effect, at the inventions of both dynamite and the submarine. Third, the model explicitly excludes domestic politics, human psychology, and normative commitments, each of

²⁹This assumes that the damage a war inflicts does not affect the returns of entering it, which is not necessarily true.

³⁰Earl C. Ravenal, "Counterforce and Alliance: The Ultimate Connection," *International Security* 6, no. 4 (Spring 1982): 26–43, muse.jhu.edu/article/446729.

which demonstrably shapes nuclear decision-making.³¹

Alternative accounts of non-use include the shadow of the future, the limited strategic utility of nuclear weapons, material constraints, the obsolescence of major war, and the nuclear taboo.³² The relevant implication is not that deterrence fails, but that non-use rests on multiple supports of uncertain strength. A civil defense initiative interacts with several of them at once, and in both directions. Whether it stabilizes or destabilizes is therefore not answerable in the abstract but instead depends on which mechanisms the intervention activates.

Hence, (4), the elasticity ε is not a constant of nature. It is assembled out of deterrence, accident dynamics, norms, and targeting doctrine, and an intervention can move its components separately. The subsections that follow take up those components one at a time— first the empirical baseline for ε , then the three mechanisms unique to nuclear warfare that shift it.

3.3 The Empirical Baseline: Severity and Frequency

The inverse relationship assumed above is a theoretical construct. Its empirical counterpart is the power law relating the severity of interstate wars to their frequency, established by Richardson and refined by Clauset.³³ In Clauset’s estimation, the severity of a war, measured in battle deaths X , is drawn from a probability density

$$p(x) = A x^{-\alpha}, \quad \alpha \approx 1.53, \quad x \geq D_{\min} = 7,061, \quad (6)$$

where A is fixed by normalization. Integrating p over its support and setting the total to one gives

$$1 = \int_{D_{\min}}^{\infty} A x^{-\alpha} dx = \frac{A}{\alpha - 1} D_{\min}^{1-\alpha} \implies A = (\alpha - 1) D_{\min}^{\alpha-1}, \quad (7)$$

which converges precisely because $\alpha > 1$; a shallower exponent would not describe a probability distribution at all.

The object that enters the expected-damage calculus, however, is not the density but the *tail*: the probability that a war, if one occurs, is at least as severe as D . Integrating the density from D upward,

$$P(X \geq D) = \int_D^{\infty} (\alpha - 1) D_{\min}^{\alpha-1} x^{-\alpha} dx = \left(\frac{D}{D_{\min}} \right)^{1-\alpha} = \left(\frac{D}{D_0} \right)^{1-\alpha}, \quad (8)$$

writing $D_0 = D_{\min}$ as a reference. This is the corollary we will invoke throughout. Namely, *the probability of a war at least as severe as D scales as $D^{1-\alpha}$* . Annualizing multiplies (8) by the (roughly constant) rate at which wars break out; a constant prefactor changes no elasticity and no conclusion below, so we suppress it. The expected damage associated with courting a war of scale D is then

$$\bar{D}(D) = D \left(\frac{D}{D_0} \right)^{1-\alpha} = D_0^{\alpha-1} D^{2-\alpha}, \quad (9)$$

³¹Nina Tannenwald, *The Nuclear Taboo: The United States and the Non-Use of Nuclear Weapons since 1945* (Cambridge: Cambridge University Press, 2007).

³²Ibid.

³³Aaron Clauset, “On the Frequency and Severity of Interstate Wars,” in *Lewis Fry Richardson: His Intellectual Legacy and Influence in the Social Sciences*, ed. Nils Petter Gleditsch (Cham: Springer, 2020), 113–27, doi.org/10.1007/978-3-030-31589-4_10.

and the empirical elasticity of war probability with respect to damage is read directly off (8):

$$\varepsilon_{\text{emp}} = -\frac{d \ln P(X \geq D)}{d \ln D} = \alpha - 1 \approx 0.53. \quad (10)$$

Figure 1 sets this machinery against the historical record itself.

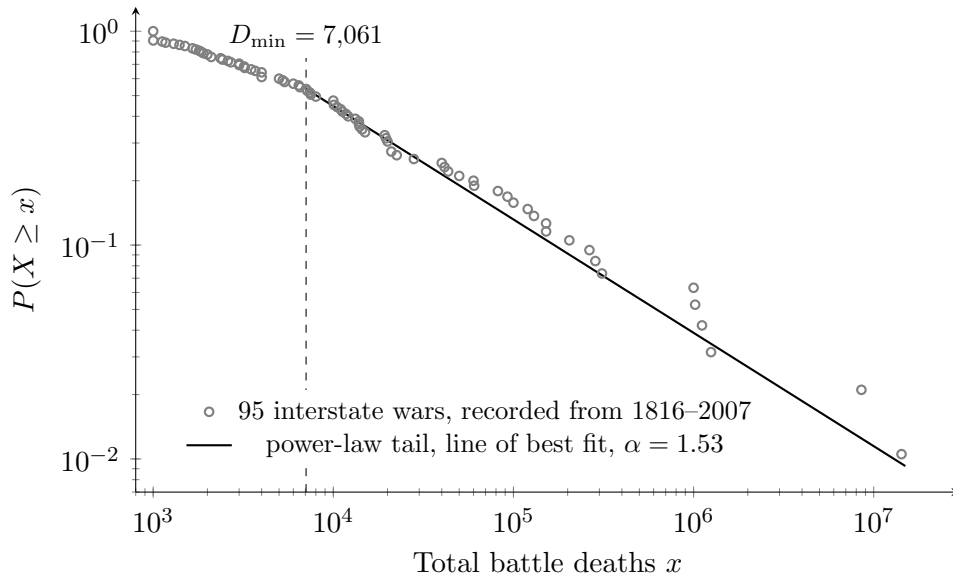


Figure 1: The empirical severity–frequency distribution of interstate war. Points constitute the cumulative distribution of total battle deaths for the 95 interstate wars that occurred between 1816–2007, recorded in the Correlates of War Inter-State War Data v4.0.³⁵ The line is the power-law tail $P(X \geq x) \propto x^{1-\alpha}$ with the maximum-likelihood fit $\alpha \approx 1.53$ above $D_{\min} = 7,061$ as reported by Clauset.

Two features of this relationship deserve comment before it is put to work. First, as Clauset observes, establishing a statistical power law through empirical observation is particularly difficult, since extremely high-impact events occur extremely infrequently, and observation therefore reflects mostly smaller-impact events. Uncertainty is highest precisely on the upper tail, the region most likely to be occupied by a nuclear exchange. Second, the law is contingent on casualties exceeding 7,061. This threshold’s precision is largely irrelevant here: the Soviet Union, upon full implementation of civil defense measures and against counterforce strikes, would still have suffered roughly four percent population loss and thus millions of deaths.³⁶ The power law is simply operating on a domain where it is expected to work. Throughout, we follow the Richardson–Clauset convention in using damage to refer to deaths, which is also what civil defense principally aims to shield.

Now compare (10) with the knife edge (5). The naive inverse relationship of 3.1 is the unit-elastic case $\varepsilon = 1$; in the parameterization of (8) that corresponds to a tail exponent of one, i.e.

³⁵Data: Meredith Reid Sarkees and Frank Wayman, *Resort to War: 1816–2007* (Washington, DC: CQ Press, 2010); Correlates of War Inter-State War Data v4.0, correlatesofwar.org/data-sets/cow-war/. Fit: Aaron Clauset, “Trends and Fluctuations in the Severity and Frequency of Interstate Wars,” *Science Advances* 4, no. 2 (2018): eaao3580, doi.org/10.1126/sciadv.aao3580; Clauset, “On the Frequency and Severity of Interstate Wars,” 2020.

³⁶Daniel Gouré, “The Tyranny of Forward Presence,” *Naval War College Review* 54, no. 3, 2001. digital-commons.usnwc.edu/cgi/viewcontent.cgi?article=2456&context=nwc-review.

$\alpha = 2$. The estimated exponent $\alpha \approx 1.53$ puts the empirical elasticity at 0.53, well below one: decreased damage does not increase the probability of war at a rate fast enough to keep up with the damage saved, and reducing damage therefore reduces expected damage. Concretely, from (8) and (9), halving the damage of a prospective war multiplies its probability by

$$\left(\frac{1}{2}\right)^{1-\alpha} = 2^{\alpha-1} = 2^{0.53} \approx 1.44, \quad (11)$$

while expected damage is multiplied by

$$\left(\frac{1}{2}\right)^{2-\alpha} = 2^{\alpha-2} = 2^{-0.47} \approx 0.72 : \quad (12)$$

the war becomes 44 percent more likely but the yearly toll falls by 28 percent. Empirically, wars sit on the favorable side of the edge. The same arithmetic connects to a classical fact about power laws: for $\alpha < 2$ the mean of the distribution (6),

$$\mathbb{E}[X] = \int_{D_{\min}}^{\infty} x p(x) dx = (\alpha - 1) D_{\min}^{\alpha-1} \int_{D_{\min}}^{\infty} x^{1-\alpha} dx, \quad (13)$$

diverges, because the integrand falls off as $x^{-0.53}$ and the integral has no finite value. Equivalently, by (9), $\bar{D}(D) \propto D^{0.47} \rightarrow \infty$ as $D \rightarrow \infty$. The heaviness of the tail therefore not only renders nuclear war as the dominant term in expected damage but also the feature that makes damage reduction a worthwhile pursuit at the baseline.

This result slightly alters our overarching question: it is no longer whether damage limitation is self-defeating in general, but whether there are factors unique to nuclear warfare, or to civil defense as a means of mitigating it, that would be expected to shift this curve. We identify four candidates. First, a more or less constant possibility of nuclear war stemming from accidents, such as false alarms leading to rapid and destructive escalation. Second, a nuclear taboo causing the probability of war to increase even more slowly with decreased damage, possibly bolstered by the cultural effects of civil defense programs. Third, an increased viability of counterforce strategies enabled by civil defense measures, raising the incentive for states to strike. Fourth, the externalities of nuclear conflict, whose damage falls outside the casualty metric accounted for by the power law at hand.³⁷ The first two favor civil defense. The third does not, and it is the only one that poses a serious threat to the case.

3.4 A Probability Floor for Accidents

The first factor is the probability of a nuclear war stemming from an accident. To give a sense of what accidents look like, we consider the history of close calls that occurred throughout the Cold War. These span from particularly well-known examples, such as a Soviet submarine commander ordering the launch of a nuclear missile upon mistaking American signaling for the outbreak of nuclear war, to less known cases, such as a US bomber crashing over Greenland and nearly triggering a nuclear explosion, which would have prompted a US response were it mistaken for a Soviet attack.³⁸ A number of factors contribute to the probability of war stemming from events such as these, from the precision of decision-making, meaning the measures taken to guard against false alarms, to human risk aversion, which naturally produces hesitance to initiate a nuclear war even under the belief that one has sufficient reason to do so.

³⁷The fourth is developed in subsection 3.6 and remains the least resolved in the research.

³⁸Eric Schlosser, *Command and Control* (New York: Penguin, 2013), 319–325.

This latter contribution is itself partly contingent on civil defense. Decision-makers may be less willing to initiate a conflict they believe will be more deadly, and if we assume that humans are risk-averse, this may follow a curve that converges faster than simple inverse proportionality. Given, however, that the damage of any nuclear war is likely to be quite large, the contribution from risk-aversion is likely to be outweighed by some static probability of nuclear accidents: truly random false alarms carrying an inevitable likelihood of escalation. Adding this contribution to the baseline yields

$$\bar{D} = D \left(\left(\frac{D}{D_0} \right)^{1-\alpha} \oplus P_a \right), \quad P_a = P_\psi(D, *) + C, \quad (14)$$

where P_ψ collects the decision-theoretic contributions (risk aversion being among them) and C is the static floor, and where $\oplus$ denotes the addition operator for probabilities: for events A and B ,

$$a \oplus b = a + b - ab, \quad \text{i.e.} \quad P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) \quad (15)$$

under independence. Two properties of $\oplus$ matter here. It never exceeds one (so probabilities remain probabilities under composition), and for low-probability events, which the outbreak of nuclear war presumably is, the cross term ab is second-order small, so $\oplus$ is approximately additive for independent contributions:

$$a \oplus b = a + b - ab \approx a + b \quad \text{when } a, b \ll 1. \quad (16)$$

Since the probabilistic variable term is much smaller than the value of C for sufficiently large D , (14) is approximately

$$\bar{D} = D \left(\left(\frac{D}{D_0} \right)^{1-\alpha} \oplus C \right). \quad (17)$$

Two caveats are worth registering. First, the variable term from human risk-aversion may grow fast enough as damage decreases to outweigh the reduction in damage. We assume it does not, on two grounds: even a relatively low-damage nuclear war is extremely dangerous and sits well within the region where risk aversion severely suppresses the probability contribution, and an individual-level nuclear taboo damps the growth of that contribution as damage decreases. Second, conventional wars have been triggered accidentally, through miscalculation, and some of this contribution may already be absorbed into the naïve power law.³⁹ We assume that nuclear wars are uniquely vulnerable to such accidents relative to historical conventional warfare and nuclear close-calls, owing to the facility with which they may be initiated and the need to respond swiftly to an apparent attack to avoid severe counterforce damage and a subsequent loss of response capabilities.

The structural consequence is significant for this research's central argument, and it is worth walking through the arithmetic that produces it. Expanding (17) with the definition of $\oplus$, we have that

$$\bar{D} = D \left[\left(\frac{D}{D_0} \right)^{1-\alpha} + C - C \left(\frac{D}{D_0} \right)^{1-\alpha} \right] = \underbrace{(1 - C) D_0^{\alpha-1} D^{2-\alpha}}_{\text{deliberate term}} + \underbrace{C D}_{\text{accident term}}. \quad (18)$$

³⁹B. Radzinsky, "Miscalculation, Misperception, and Risk Reduction," *Lawrence Livermore National Laboratory*, LLNL-TR-826600, Sept 2021. osti.gov/servlets/purl/1820013.

Both terms are increasing in D , as differentiating yields

$$\frac{d\bar{D}}{dD} = (1 - C)(2 - \alpha) D_0^{\alpha-1} D^{1-\alpha} + C > 0 \quad \text{for } \alpha < 2, \quad (19)$$

so expected damage rises monotonically with damage everywhere on the domain, and the accident term grows without bound: a constant probability floor does not scale down as damage is reduced. A probability cannot be subtracted; expected damage diverges at infinity. That is, as the damage of a potential war grows, the accident term keeps its probability from falling fast enough to compensate, meaning that reducing damage is unambiguously worth it and the paradox does not bind under this calculus. Whatever the elasticity of the deliberate term, the floor contributes an elasticity of zero. That is, no reduction in damage buys any reduction in the rate of false alarms, meaning that no term could possibly counteract the reduction in expected damages via an increase in probability.

Figure 2 shows the shift on the probability curve itself.

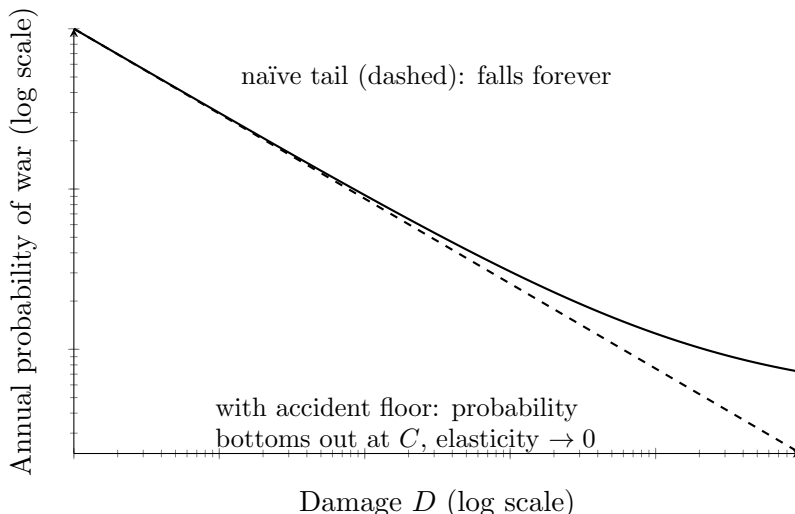


Figure 2: The accident floor. The naïve tail (dashed) falls without limit as damage grows; composing it with a constant accident probability C (solid, via $\oplus$) leaves the curve unchanged where deliberate war dominates but pins the probability at C .

3.5 The Nuclear Taboo: Damping the Damage Signal

The second factor unique to nuclear warfare is Nina Tannenwald's nuclear taboo. Tannenwald argues that the traditional account of nuclear non-use, deterrence through MAD, which in this framework corresponds to extremely low probability at extremely high damage on the naïve curve, is unsuccessful at making historical predictions. It fails, for instance, to account for the United States declining to use nuclear weapons against the Soviet Union during a window in the early 1950s when the power asymmetry would have made a first strike relatively safe and total. Tannenwald argues that this ineffective deterrence force takes the form of a taboo: certain qualitative feature(s) of nuclear weapons, likely their sublimity and extreme destructive capacity, create a state inclination against their use, independent of a rational assessment of the damage they would inflict. This inclination may be shaped by domestic politics, human and cognitive

psychology, or normative foreign policy commitments.⁴⁰ The effect is a damping of the influence that changes in damage exert on the probability of war. This is intangible and impossible to precisely quantify, but it plausibly manifests as a transformation applied to the naïve probability curve:

$$T : P(D) \mapsto N(1 - A) \int_D^{D_{\max}} P(D') \phi_D(D') dD', \quad \phi_D(D') \geq 0, \quad \int_D^{D_{\max}} \phi_D(D') dD' = 1, \quad (20)$$

where $D_{\max}$ is the largest damage the arsenals in play can inflict.

We begin with explaining the integral, because it is where the taboo’s defining feature — indifference to the precise damage figure — enters the mathematics. A state whose aversion attaches to nuclear war as a category does not read the probability curve at the exact value D ; it forms its inclination by weighing the contemplated war against the class of nuclear wars it could become. The weighting function ϕ_D makes that averaging explicit. Its support encodes the direction of the taboo’s imagination: a state contemplating a war of size D weighs it against the larger wars escalation could produce, never against the smaller war it might luckily turn out to be, so ϕ_D vanishes below D ; and escalation is bounded by what the arsenals can physically deliver, so ϕ_D vanishes above $D_{\max}$. Between those limits, the natural shape follows from the empirical structure of Section 3.3. War severities are distributed as a power law, which is scale-free: the ratio by which a war might grow carries no preferred size. A state that imagines escalation in the same terms imagines escalation *multiplies* D'/D that are themselves power-law distributed. We therefore take the weighting function to be a Pareto distribution in D' with scale D and shape β , truncated at the arsenal cap:

$$\phi_D(D') = \frac{\beta D^\beta D'^{-\beta-1}}{1 - (D/D_{\max})^\beta}, \quad D \leq D' \leq D_{\max}. \quad (21)$$

The denominator is the normalization, since $\int_D^{D_{\max}} \beta D^\beta D'^{-\beta-1} dD' = 1 - (D/D_{\max})^\beta$. The shape parameter β plays the role a width would play in an additive kernel, in inverse: $1/\beta$ is the spread of the state’s concern in log-damage, so a small β is a decision-maker who registers “nuclear war” and imagines it growing by large multiples, while $\beta \rightarrow \infty$ collapses ϕ_D onto the point $D' = D$, a decision-maker who parses five million deaths against eight million and averages nothing. Now, moving to the constants: the coefficient A , bounded between zero and one, represents a principled aversion to nuclear war, which lowers the chance of war across the domain when coupled with the averaging of the probability curve. N , also bounded between zero and one, is a function of damage representing normative constraints that emerge under the prospect of inflicting heavy damages; N and A serve a similar purpose of reducing the transformed probability on the basis of a nuclear war’s qualitative features.

The properties of T that carry the argument can now be derived rather than asserted.

Proposition 3.1 (The taboo cannot make war more likely). *If P is nonincreasing, then $T[P](D) \leq N(1 - A)P(D) \leq P(D)$ for every D , for any weighting function supported on $[D, D_{\max}]$.*

(The proof appears in Appendix A.)

This is not a technicality; it is the statement that a taboo is a taboo. A transformation meant to represent an inhibition on use should be unable to increase the probability of nuclear war

⁴⁰Nina Tannenwald, *The Nuclear Taboo: The United States and the Non-Use of Nuclear Weapons since 1945* (Cambridge: Cambridge University Press, 2007).

for *any* pre-existing curve, including one that is strongly convex — and the one-sided support is exactly the condition that guarantees it. (A weighting that reached below D would mix in smaller wars, where the naïve curve is higher; on a curve as convex as (8), which climbs steeply as damage falls toward zero, that average can exceed $P(D)$, turning the “taboo” into an amplifier.)

Proposition 3.2 (The taboo damps the damage signal). *Apply T with the kernel (21) to the empirical tail $P(D) = (D/D_0)^{1-\alpha}$, and write the result as $T[P](D) = N(1-A)c(D)P(D)$, where c is the attenuation produced by the averaging alone. Then, with $t = D/D_{\max} \in (0, 1]$,*

$$c(D) = \frac{\beta}{\alpha + \beta - 1} \cdot \frac{1 - t^{\alpha+\beta-1}}{1 - t^\beta}, \quad (22)$$

c is increasing in D , with $c \rightarrow \beta/(\alpha + \beta - 1) < 1$ as $D/D_{\max} \rightarrow 0$ and $c \rightarrow 1$ as $D \rightarrow D_{\max}$; and the transformed curve is everywhere flatter than the naïve one:

$$\left| \frac{d \ln T[P]}{d \ln D} \right| = (\alpha - 1) - \frac{d \ln c}{d \ln D} < \alpha - 1. \quad (23)$$

(The proof appears in Appendix A.)

The impact of this transformation is therefore a smoothing of $P(D)$: a curve that is lowered most at low damage, where escalation has the farthest to run and c sits near its floor of $\beta/(\alpha + \beta - 1)$, and scarcely at all as damage approaches the arsenal cap, where there is nowhere left to escalate and c is near one, with a further uniform reduction from the N and $(1 - A)$ coefficients. For orientation, $\beta = 1$ with $\alpha = 1.53$ gives a floor of $1/1.53 \approx 0.65$: a small nuclear war is treated as roughly two-thirds as likely as the naïve curve says, a war near the cap as exactly as likely. The transformed curve is flatter than the naïve one: the probability of war responds less to a given change in damage, which is the damping the taboo is meant to represent, and the case for civil defense strengthens accordingly. In elasticity terms, the taboo pushes the effective ε *below* the empirical 0.53, moving the curve further from the knife edge at one. Composing with the accident floor of 3.4, the working model becomes

$$\bar{D} = D \left(T \left[\left(\frac{D}{D_0} \right)^{1-\alpha} \right] \oplus C \right). \quad (24)$$

Figures 3 and 4 separate the transformation’s two effects on the curve.

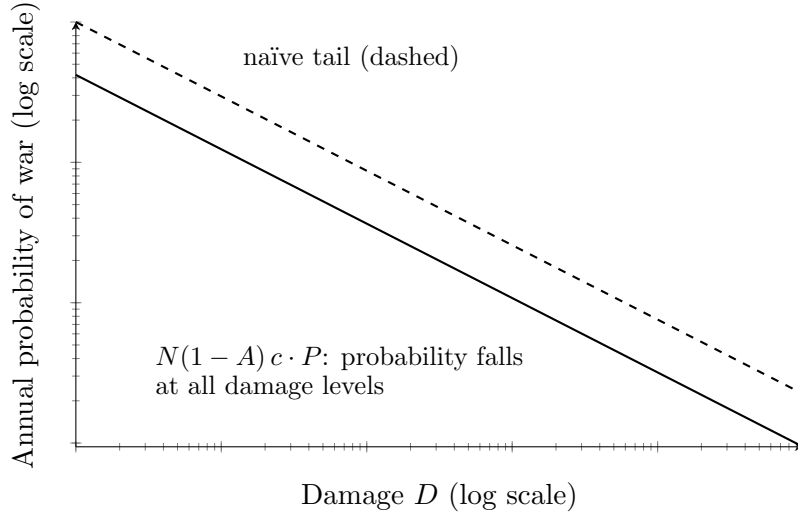


Figure 3: Taboo effect one: aversion to nuclear war by virtue of its qualitative features. The coefficients $N(1 - A)$ and the attenuation constant $\beta/(\alpha + \beta - 1)$ of the untruncated kernel multiply the curve by some constant value less than one. On log axes this is a parallel downward shift: war is less likely at every damage level, while the elasticity influencing the paradox calculus remains untouched.

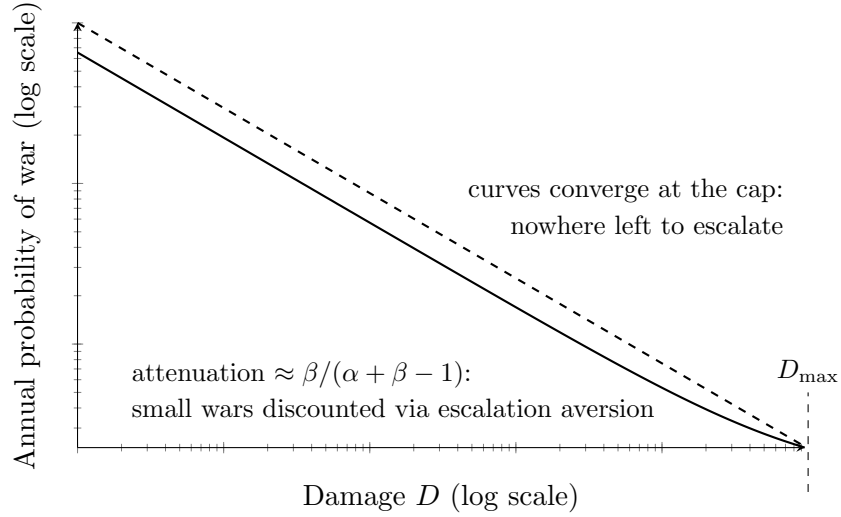


Figure 4: Taboo effect two: dampening from the truncated escalation kernel ($\beta = 1$, $\alpha = 1.53$). That is, the averaging suppresses small wars most, where imagined escalation has the farthest to run and vanishes as damage approaches the arsenal cap ($c \rightarrow 1$). The transformed curve is flatter than the naïve one: the probability of war responds less to a given change in damage, the dampening effect explained via Proposition 3.2.

Whether the transformation itself responds to civil defense is a question that requires further consideration, and the answer runs in both directions. Historical civil defense programs helped keep the risk of nuclear war proximate on the public's consciousness, producing a cultural

stigmatization of nuclear weapons that could manifest as increased state aversion and thus a higher A . In contrast, civil defense could reduce the extremity of nuclear war, as in the Soviet case where a successful civil-defense-backed counterforce strike could at one point have brought casualties beneath the levels of conventional warfare, leading to a more rational assessment of the costs and benefits of nuclear use, likely leading to attention to actual damage. That would narrow the weighting function (i.e. raise β) and could also lower A .

This research suggests that A has increased over time, which is in turn compatible with a narrower weighting function. The Cold War public exhibited extreme risk-aversion to the prospect of a nuclear war but parsed more finely on expected damage within an accepted probability of nuclear war. The cultural record reviewed in Section 2 supports the first half of this argument, as civil defense made the risk of nuclear war feel both imminent and proximate to the American public, and the aversion it produced was not calibrated for technical parameters of any specific scenario.⁴¹

Two constraints follow. First, provided that civil defense does not break the taboo outright, any reasonable degree of smoothing works in its favor. That is, even where civil defense narrows the weighting function, that unsmoothing cannot, at high damage values, undo the benefit of reduced damage, since by Proposition 3.2 the effect of smoothing diminishes as damage rises. The bound is explicit: raising β can lift the probability at any damage level by at most the factor $1/c(D)$, which is at most $(\alpha + \beta - 1)/\beta$ far from the cap and tends to one near it, so there is progressively less smoothing left to undo exactly where nuclear wars sit. Second, a taboo should be unable to increase the probability of nuclear war. This requires that the weighting function vanish below D , so that the transformation cannot raise the probability of any pre-existing curve, including one that is strongly convex — the content of Proposition 3.1.⁴²

One implication deserves noting: holding A constant with respect to damage, and therefore with respect to civil defense, the taboo contribution does not change the divergence properties of the expected damage function. This too is now a calculation rather than a claim: by Proposition 3.2, c is bounded between $\beta/(\alpha + \beta - 1)$ and one on the whole domain, so $T[P]$ lies within constant factors of P and the transformed expected damage retains the exponent $2 - \alpha$ of (9). Depending on how the weighting function is constructed, however, radically different behavior could emerge, and two cases fix the range. If the arsenal cap is removed — the untruncated Pareto, $D_{\max} \rightarrow \infty$ — then $t \rightarrow 0$ for every D and c collapses to the constant $\beta/(\alpha + \beta - 1)$: the averaging becomes scale-free, the taboo reduces to a uniform discount indistinguishable from $(1 - A)$, and no flattening occurs at any scale. The damping of Proposition 3.2 is therefore not a property of the taboo alone but of the taboo operating inside a finite arsenal; it is the cap that gives escalation an absolute limit and thereby makes small wars more heavily discounted

⁴¹See Masco, *The Future of Fallout*; Garrett, “Civil Defense and the Public.” Garrett’s compilation documents the knowledge-action gap, as respondents who regarded fallout as a major danger reported not knowing what to do about it. Notably, this aversion specifically pertained to the categorization of nuclear war rather than the parameters of any particular exchange.

⁴²This Pareto form follows the literature on statistically modeling war. Richardson first established that war sizes follow a power law; Cederman shows the distribution arising from scale-invariant escalation dynamics, and Braumoeller argues that escalation is approximately scale-independent, which is precisely the process this weighting function represents. On the Pareto distribution and power laws generally, see Newman. Lewis Fry Richardson, “Variation of the Frequency of Fatal Quarrels with Magnitude,” *Journal of the American Statistical Association* 43, no. 244 (1948): 523–546; Lars-Erik Cederman, “Modeling the Size of Wars: From Billiard Balls to Sandpiles,” *American Political Science Review* 97, no. 1 (2003): 135–150; Bear F. Braumoeller, *Only the Dead: The Persistence of War in the Modern Age* (New York: Oxford University Press, 2019); M. E. J. Newman, “Power Laws, Pareto Distributions and Zipf’s Law,” *Contemporary Physics* 46, no. 5 (2005): 323–351.

than large ones. If instead the shape parameter varies with damage, the direction of the effect follows from (22): a β that *falls* with D — a state imagining ever-larger escalation multiples at larger scales — lowers c as D grows and steepens the curve, while a β that rises with D flattens it further. The adage that one death is a tragedy and one million is a statistic gestures at the former: numbing that widens with scale makes the state more, not less, averse to the largest wars, converting the power law into something steeper than a power law of logarithms at its upper end. The shape of the weighting function may be strongly damage-dependent, producing non-obvious changes.

3.6 Counterforce as a Binding Mechanism

The first two factors strengthen the case for civil defense. The third is the only one that challenges it: civil defense may raise the risk of nuclear war by enabling a counterforce-centric strategy. Broadly, a counterforce strategy uses nuclear weapons not to hold cities and economies hostage but to destroy an adversary’s nuclear and military capabilities, leaving the initiator in a stronger position to negotiate after the strike. Its corollary is a strong incentive to strike first. Under MAD, the order of strikes is irrelevant, because both sides are destroyed regardless. However, under counterforce, striking first confers an advantage, and a state that holds it becomes, at the very least, more trigger-happy. Hence, counterforce and a first strike capability are mutually dependent: a first strike with any rational motive can only be justified as counterforce, and a counterforce strike is only fully effective if it comes first.

Meanwhile, civil defense, by definition, protects the public and the ordinary functioning of society that countervalue targeting destroys and counterforce mostly spares. Though counterforce strikes do not always avoid population centers, with some military targets lying inside cities, civil defense enters the counterforce decision by a different route. A state contemplating a disarming first strike must expect to absorb whatever surviving retaliatory measures exist, much of which will ultimately inflict damages on civilians and nonmilitary infrastructure. Naturally, civil defense makes this residual more tolerable, entering the counterforce calculus not as a defense against counterforce but as a defense against the reprisal that such a form of attack invites.

This mechanism is not confined to nuclear weapons. Conventional precision campaigns against an adversary’s strike capability follow the same logic.⁴³ They too are launched expecting retaliation against cities, and where the target can reach the attacker’s population, the attacker’s own preparedness helps make the campaign sustainable. Israel’s 2025 campaign against Iran’s nuclear and missile infrastructure offers a clear recent example. Retaliation against Israeli civilians was expected from the first hours, and an Israeli home-front emergency was subsequently declared, with sheltering significantly reducing Israeli civilian casualties.⁴⁴

This dynamic suggests that counterforce warrants some treatment within the model. The

⁴³Keir Lieber & Daryl Press, “The New Era of Counterforce,” *International Security* 41(4), 2017.

⁴⁴Regarding the strategic logic underlining Israel’s campaign against Iran’s missile forces, see Fabian Hinz, “Iran’s Shattered Missile Strategy,” *Survival* 67, no. 4 (2025), doi.org/10.1080/00396338.2025.2534280; and Sam Lair, “Shallow Ramparts: Air and Missile Defenses in the June 2025 Israel-Iran War,” Foreign Policy Research Institute, October 17, 2025, fpri.org/article/2025/10/shallow-ramparts-air-and-missile-defenses-in-the-june-2025-israel-iran-war/. Regarding the home-front declaration, see Amichai Cohen and Mirit Lavi, “Operation Rising Lion: A Special Home Front Situation,” Israel Democracy Institute, March 2, 2026, en.idi.org.il/articles/63607. Regarding sheltering and casualties, the Home Front Command assessed that reinforced spaces “saved many lives,” with occupants of shelters directly adjacent to impact sites unharmed; see Emanuel Fabian, “IDF Reiterates Safe Rooms Still Best Option Against Iran’s Missiles, Despite 2 Fatalities,” *Times of Israel*, June 16, 2025.

record from which the power law is estimated contains almost no cases in which protecting the population was a precondition for attempting to disarm an adversary's entire strategic capacity. Under counterforce, the probability of war responds to protected damage d rather than unprotected damage D , and a curve fitted to countervalue outcomes misses such a response. We therefore introduce F , the share of a state's expectation that a war will primarily follow counterforce rather than countervalue logic, itself a function of the protection ratio d/D :

$$\bar{D} = D \left((1 - F) T \left[\left(\frac{D}{D_0} \right)^{1-\alpha} \right] \oplus F Y(d, *) \oplus C \right). \quad (25)$$

Here Y is the probability of nuclear war under counterforce logic, a function of d , the damage sustained *with* civil defense, rather than D , the damage without it. The two must be kept apart to isolate the effect of such a policy measure, and the corresponding ratio of protected to unprotected damage is also precisely what pushes F upward. To make the structure explicit, write the protection ratio as

$$r = \frac{d}{D} \in (0, 1], \quad F = F(r), \quad F'(r) < 0, \quad \frac{\partial Y}{\partial d} < 0. \quad (26)$$

These equations respectively mean that deeper protection (a smaller r) makes counterforce logic a larger share of a state's expectation. Naturally, a disarming strike is conceivable precisely to the extent that a retaliatory act is survivable. If Y depended on D while F depended on d/D , then protection would enter the same expression twice through one variable and the model would feed on itself. Splitting the arguments as in (25) prevents such an issue.

With effective civil defense, Y still falls as damage rises, but the product FY can rise even as d falls. This is the concern for civil defense. Differentiating the counterforce term along a deepening of protection (d decreasing, D held fixed), we get

$$\frac{\partial(FY)}{\partial d} = \underbrace{F'(r) \frac{1}{D} Y}_{<0} + \underbrace{F \frac{\partial Y}{\partial d}}_{<0} < 0, \quad (27)$$

such the counterforce contribution to war probability strictly *increases* as protection deepens. Both channels push the same way, the direct one (a smaller reprisal is easier to accept) and the compositional one (a smaller reprisal shifts expectation toward the strategy for which reprisal size is the binding constraint). The same decomposition shows when the product can rise with damage. That is, moving along a path on which d and D both grow but protection deepens in ratio terms (r falling), the increase in F can outweigh the decline in Y whenever the elasticity of F with respect to r exceeds the elasticity of Y with respect to d , a very possible case.

Figure 5 shows the resulting shift.

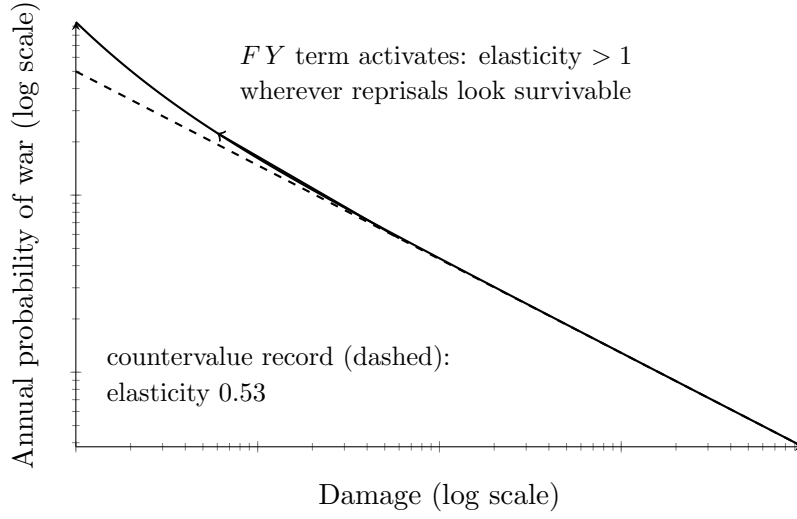


Figure 5: The counterforce shift. The historical curve (dashed) is fitted to account for counter-value outcomes. As protection deepens and the expected reprisal shrinks, the counterforce term FY of (25) activates — both F and Y rise as d falls, per (27) — and the composite curve bends upward at low damage with an elasticity exceeding one. Civil defense moves a state leftward along this curve, meaning that counterforce emphasis might push the calculus past the knife edge.

The historical record gives some credence for arriving at a specific value for F . Put simply, the United States largely pursued countervalue-based punishment while the Soviet Union sought a counterforce-based means of undermining U.S. capabilities (i.e. denial).⁴⁵ Of course, neither party was sworn to one specific strategy: the United States, especially under Reagan, did take an interest in winning a nuclear war through civil defense, in defiance of MAD, and the Soviet Union understood the incentive-structure produced by countervalue targeting. A national strategy can therefore be partly counterforce, but given how sharply the two stances differed, F appears to jump with certain variables rather than vary smoothly.

We argue that F tends toward zero over time for two reasons: faster information flow and faster delivery. First and foremost, a state launching a counterforce strike wants a minimum level of civilian and economic protection in place. It must achieve that within a window that opens with the first detectable signs of launch preparation or civil defense, such as evacuation or mass sheltering, and closes with the arrival of an enemy strike. Writing τ_{prep} for the time protection takes, τ_{detect} for the time until the adversary detects it, and τ_{strike} for the transit time of the adversary's response, the strategy is feasible only if

$$\tau_{\text{prep}} \leq \tau_{\text{detect}} + \tau_{\text{strike}}, \quad (28)$$

and protecting millions of people by any practical means naturally takes time. Indeed, such an endeavor would likely take a lot more time than it takes adversarial response capacity to materialize with the increased velocity of the flow of information (shrinking τ_{detect}) and missile capacity

⁴⁵Fritz W. Ermarth, "Contrasts in American and Soviet Strategic Thought," *International Security* 3, no. 2 (Fall 1978): 138–155; Richard Pipes, "Why the Soviet Union Thinks It Could Fight and Win a Nuclear War," *Commentary* 64, no. 1 (July 1977): 21–34.

(shrinking τ_{strike}), suggesting that the strategy gradually becomes less and less feasible.⁴⁶

This fact has two implications for civil defense, and the second is the more troubling. First, adopting civil defense measures signals a sharply elevated probability of attack. Hence, the advanced sheltering of populations during periods of diplomatic tension is an unreasonable and dangerous strategy that would sharply escalate tensions. Soviet practice demonstrates such an issue: the program shifted toward shelter-in-place precisely because evacuation, which takes longer to implement, signals intent, and Americans read the program as first-strike preparation. The resilience literature independently reproduces the same Cold War worry: Korhonen notes that American hawks read sophisticated Soviet civil defense as evidence the Soviet Union was preparing to initiate nuclear war, a perception that could itself raise escalation risk in a crisis.⁴⁷ The paradox this paper formalizes is therefore not an artifact of Cold War strategic culture but instead recurs wherever preparedness is proposed. Second, the dangers of counterforce disappear only when certain civil defense measures stop working. Any measure must be carried out within a window set partly by how early warning arrives, and the defender always has less warning than the attacker. In the notation of (28), the defender’s window is τ_{strike} alone while the attacker’s includes its own preparation lead. Hence, any measure the attacker cannot perform, the defender naturally cannot perform either. This matters little for counterforce strikes, which do not aim at civilians. However, the blast effects of countervalue strikes, by now the only viable kind, become particularly hard (i.e. virtually impossible) to defend against. This is no sign of retreat on this paper’s goal of providing implementable policy measures. Rather, it suggests that investment should favor the kinds of civil defense that stand the test of time. Evacuation and shelter programs are dangerous precisely when they work and subsequently enable counterforce strikes.

One last consideration falls outside the three factors and remains unresolved. The power law counts damage in casualties, assuming that economic and social damage scale in proportion, with states caring about both. That assumption is likely false under sophisticated civil defense, which protects civilians far better than infrastructure. This model may therefore underestimate the extent to which civil defense reduces aggregate damages.

3.7 The Paradox as a Design Constraint

The three factors do not resolve into a single verdict. Each one alters the expected damage curve toward a different shape, while the verdict on civil defense’s effect on such a curve depends on which of the three regimes actually dominates. It is useful to fix the classification before drawing the pictures. Write the composite expected damage function of (25) generically as $\bar{D}(D) = D \cdot \Pi(D)$, where Π is the (composed) annual probability term, and classify by the local elasticity $\varepsilon(D) = -d \ln \Pi / d \ln D$ familiar from (5):

$$\frac{d \ln \bar{D}}{d \ln D} = 1 - \varepsilon(D) \quad \implies \quad \begin{cases} \varepsilon < 1 \text{ everywhere} & \text{regime one: } \bar{D} \text{ increasing,} \\ \varepsilon > 1 \text{ everywhere} & \text{regime two: } \bar{D} \text{ decreasing,} \\ \varepsilon \text{ crosses 1} & \text{regime three: } \bar{D} \text{ is U-shaped.} \end{cases} \quad (29)$$

⁴⁶Keir A. Lieber and Daryl G. Press, “The New Era of Counterforce: Technological Change and the Future of Nuclear Deterrence,” *International Security* 41, no. 4 (Spring 2017): 9–49.

⁴⁷Janne M. Korhonen, “Sheltering humanity against x-risk: report from the SHELTER weekend,” Effective Altruism Forum, Oct. 2022. forum.effectivealtruism.org/posts/DArtSnDxRH5AsE5RF/sheltering-humanity-against-x-risk-report-from-the-shelter.

In the first, expected damage grows without bound as wars get larger, and the probability falls too slowly to offset the rising damage. Hence, the larger the war, the worse the effect on human welfare, and reducing damage through civil defense is necessarily worth it. The accident floor and the taboo’s dampening effect both push toward this regime: the floor contributes elasticity zero (Section 3.4), and the taboo pushes the deliberate term’s elasticity below its empirical value of 0.53 (Proposition 3.2), so the composite elasticity sits below one across the domain and $\bar{D}$ rises monotonically, dominated at scale by the accident term CD of (18). Visually, this looks like:

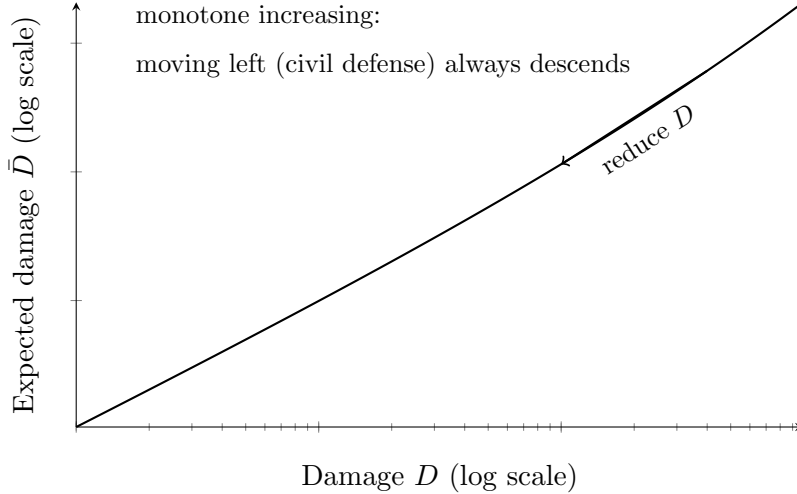


Figure 6: Regime one. With the accident floor and the damped deliberate term, the probability of war falls more slowly than damage grows ($\varepsilon < 1$), so expected damage rises without bound as wars get larger. Any reduction in damage moves down the curve.

In the second, the relationship is inverted: expected damage grows as wars get smaller. Such a shift results from the probability of the war being more elastic to how “survivable” the war appears. Intuitively, this shift means that halving the damage of a given war more than doubles its probability of occurrence ($\varepsilon > 1$). Hence, plotted against damage, expected damage falls monotonically. Under such a regime, the curve has no interior minimum, and expected damage falls all the way out to the largest possible war in light of probability collapsing faster than damage grows: the formal optimum sits at the arsenal cap, the boundary of the feasible domain, and every increment of protection is a loss. This is the logic of MAD as good for humanity, and the shift from countervalue to counterforce pushes toward this regime, because by (27) a disarming first strike becomes thinkable exactly when the striker believes its population can ride out the reprisal. Visually, this looks like:

In the third, both effects operate at once, and expected damage is more serious at both extremes such that the curve resembles a U, and reducing expected harm becomes a matter of balancing the relative decrease in damages that would follow from civil defense with the relative increase in probability that naturally follows. The two arms are analytically distinct contributions. A minimal composite exhibiting both is

$$\bar{D}(D) = a D^{1-\eta} + C D, \quad \eta > 1, \quad (30)$$

where the first term is a deliberate-war contribution whose probability is more than unit-elastic

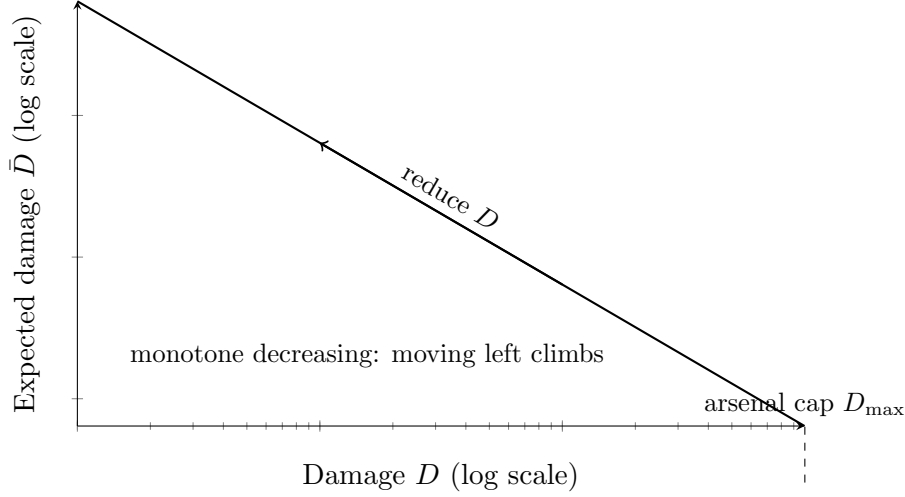


Figure 7: Regime two. When probability is more than unit-elastic in survivable damage ($\varepsilon > 1$), expected damage falls monotonically in D : the curve has no interior minimum, the optimum sits at the boundary $D_{\max}$, and every reduction in damage climbs the curve.

(the counterforce channel, elasticity η) and the second is the accident term of (18). The first term diverges as $D \rightarrow 0$; the second diverges as $D \rightarrow \infty$; and the interior minimum follows from the first-order condition

$$\frac{d\bar{D}}{dD} = (1 - \eta) a D^{-\eta} + C = 0 \implies D^* = \left(\frac{(\eta - 1) a}{C} \right)^{1/\eta}, \quad (31)$$

with the second derivative $\eta(\eta - 1)aD^{-\eta-1} > 0$ confirming a minimum. Visually, this looks like:

The viability of counterforce raises this left arm: deep protection makes war thinkable. Meanwhile, large-scale accidents, precisely the mechanism that civil defense instrumentally reduces, raise the right portion, as false alarms occur at a fixed rate, and an accident at maximal damage naturally yields the worst expected outcome. It is the fundamental goal of civil defense to aid the U.S. in lying at the minimum of the expected damage curve.⁴⁸

This is not an easy optimum to find. Very similar interventions can ultimately rest in different regimes depending on specific characteristics, such that the key question becomes which specific measures constituting civil defense reduce expected damages under the specific set of conditions at hand.

Four conditioning factors emerge from the mechanisms above. First, considering whether or not the intervention is observable is key, as an adversary can only account for what it can see in forming its targeting calculus. Second, can the measure be surged in a crisis? A capability that can be activated during a confrontation signals intent and compresses the adversary's decision timeline, the dynamic emphasized during discussion of elevating the counterforce window in 3.6. The third is what the intervention protects. Measures that preserve strike capability or government continuity directly influence an adversary's counterforce calculus, while measures that protect recovery capacity, such as food systems and trade relationships, do not enter such a calculus at all. The fourth is the taboo channel engaged by the intervention. Those embedded

⁴⁸Divergence is used loosely here; with a finite arsenal there is a maximum damage value, and probability cannot exceed unity, so neither limit is literally attained.

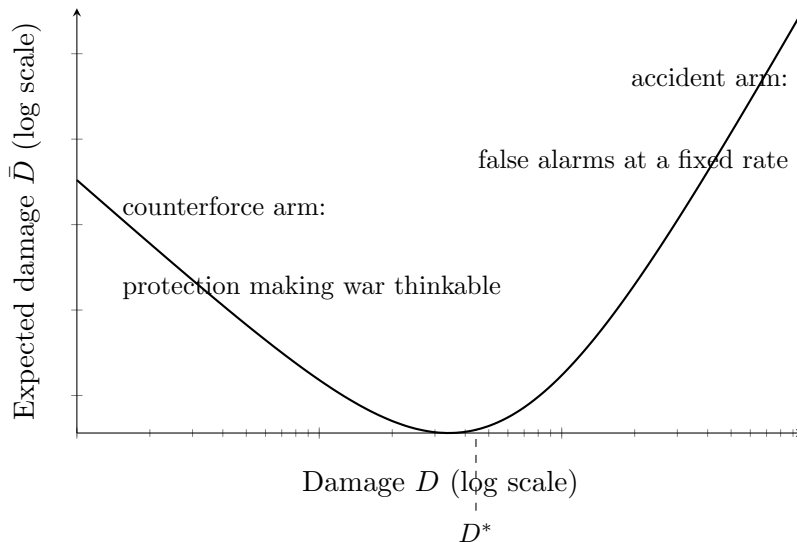


Figure 8: Regime three. Superposing the counterforce shift (left arm, diverging as $D \rightarrow 0$) and the accident floor (right arm, diverging as $D \rightarrow \infty$) produces a U with interior minimum D^* given by (31). Some protection is worth having and more is not: civil defense can overshoot D^* .

in ordinary civic and economic life strengthen the prohibition on use. Meanwhile, spectacular, fear-forward infrastructure erodes it.

Harvey Averch reached a similar conditioning logic during the Cold War. His 1963 memorandum took as initial conditions that “neither side currently has more than a modest civil defense program nor has made any extensive preparations for postattack reorganization and recuperation,” and treated civil defense as “a necessary condition for a damage-limiting capability” whose meaning depended entirely on strategic context. In his Case I, the Soviet Union seeks strategic superiority while the United States seeks stable deterrence. There a robust Soviet program is essential to superiority, since damage limitation requires protecting the population and economy against the U.S. retaliatory strike. This same program could also serve as a lure, with the Soviets honoring agreements until new weapons were ready and adequate civil defense measures had blunted the penalty before repealing such agreements. In his Case II, both sides seek stable deterrence, and the identical Soviet program reads as prudent insurance against accidental war and first strikes. Hence, the existence of a civil defense program alone supports no analytical conclusion about an adversary’s strategy.⁴⁹

However, it follows that different types of civil defense indeed have different effects on the countervalue–counterforce balance, and more generally on each mechanism identified above. This observation transforms the paradox regarding the characteristics of interventions and their

⁴⁹Harvey A. Averch, *Strategic Ambiguity, Asymmetry and Arms Control: Some Basic Considerations*, RM-3426-PR (Santa Monica, CA: RAND Corporation, March 1963), apps.dtic.mil/sti/pdfs/AD0404010.pdf. Averch’s exact phrase is “a necessary condition for *the Soviet Union’s* damage-limiting capability”: his claim attaches to the Soviet program specifically, though the conditioning logic he applies to it is general. See also U.S. Congress, Joint Committee on Defense Production, *Civil Preparedness Review, Part 1: Emergency Preparedness and Industrial Mobilization*, report (Chairman William Proxmire), February 1977, babel.hathitrust.org/cgi/pt?id=umn.31951002826765f&seq=7; and Bernard Brodie, *The Absolute Weapon: Atomic Power and World Order* (New Haven, CT: Yale Institute of International Studies, February 15, 1946), osti.gov/opennet/servlets/purl/16380564-wvLB09/16380564.pdf.

varying effects into a design constraint. An intervention that is invisible to adversary targeting, cannot be surged in a crisis, protects recovery rather than strike capability, and lives inside an ordinary institutional makeup alters strictly the damage term within the expected value calculation while leaving the probability term essentially untouched. In the terms developed above, it enters D without entering F , and the paradox passes over it: it moves $\bar{D}$ down the curve of Figure 6 without steepening the left arm of Figure 8. Though we will argue that the traditional Cold War civil defense portfolio fails every one of these criteria in Section 5, the re-conceptualized portfolio developed in Section 6 is selected on this basis of avoiding the paradox.

4 Methodology

The paradox developed in the previous section makes the evaluation of civil defense a problem of decision-making under uncertainty: the quantities that determine an intervention’s expected value are real but resistant to point estimation. Our methodology is built specifically in light of this condition.

First, we note what we believe these probabilities should be used for. It is well understood that predictions are often wrong, as is their nature. Hence, quantitative prediction tools can be counterproductive when decisionmakers either relentlessly argue about assumptions or unflinchingly accept predictions with overconfidence. Conditions of deep uncertainty also arise when decisionmakers cannot directly know the relationship between actions and outcomes, as in civil defense, where we cannot conduct “practice runs” under simulations of real conditions to gauge the potential for public panic, or which model to use.⁵⁰ Thus, we argue for a form of robust decision making, whereby we determine how a plan might fail across many plausible futures, and how such failures can be reduced.⁵¹ Hence, it is beneficial to explore combinations of assumptions in observing how candidate policies perform, which in turn requires clearly defining the options at-hand and the threshold determining which policies are “good.” Similar to Section 3, we will proceed under a notion of “risk” defined by the expected damage to infrastructure and human welfare.

4.1 Defining Acceptable Risk

Magnitude-weighting requires a defense, because the intuition that catastrophic outcomes warrant more than proportional weight is not obviously shared by the people on whose behalf the weighting would be done. Treich and Rheinberger consider one risk to be more catastrophic than another when it is a mean-preserving spread of the less risky alternative. That is: holding expected deaths fixed, it produces more extreme outcomes and fewer moderate ones.⁵² Empirically, humans are risk-averse when facing the prospect of gain and risk-seeking when facing the prospect of a loss. In terms of a nuclear altercation, this fact means that subjects tend to prefer

⁵⁰“Making Good Decisions Without Predictions Robust Decision Making for Planning Under Deep Uncertainty.” *RAND Corporation*. (Research Highlights). Accessed Feb. 2026.

⁵¹Robert J. Lempert, et al. “Shaping the Next One Hundred Years: New Methods for Quantitative, Long-Term Policy Analysis.” *RAND Corporation*. (Report) 2003. rand.org/pubs/monograph_reports/MR1626.html

⁵²Carole Bernard, Christoph M. Rheinberger, and Nicolas Treich, “Catastrophe Aversion and Risk Equity in an Interdependent World,” *Journal of Management Science*, 64, no. 10, 13 Nov 2017. doi.org/10.1287/mnsc.2017.2859

probabilistic catastrophe over the certainty of a less extreme loss.⁵³

Two considerations keep this fact from undermining the standard adopted here. First, these preferences are category-dependent, and nuclear events sit at the extreme of the categories that elicit aversion. The mechanisms that suppress aversion elsewhere — psychological numbing, salience and memory sampling, and the breakdown of probability weighting when outcomes become abstract or incomprehensible — are reasons to distrust revealed preference over harms of this magnitude rather than defer to it. Second, policymakers may be more catastrophe-averse than voters, and in this domain they demonstrably are. Specifically, the F–N curve governing societal risk criteria in nuclear safety encodes more-than-proportional penalties for high-fatality events by rule. That precedent is imperfect. The criteria respond in part to the political fact that catastrophes produce regulatory overhauls and institutional blame, and acceptable risk can be benchmarked for reactors in a way it cannot be for nuclear war. Nonetheless, it establishes that the weighting applied here is the existing practice of the regulatory status quo.

4.2 Cost-Benefit Analysis

In performing a cost-benefit analysis, it is important to consider that voters don’t gravitate towards resilience spending for future disasters of a given probability. This research accounts for these considerations, where the general public and decision-makers alike tend to discount small probabilities and sudden-onset events that aren’t proximate.⁵⁴ Risks should not be dismissed merely because they fall below some arbitrary probability threshold, but implicit probabilities cannot be ignored; deep uncertainty cannot function as a practical policy category. Nor can precaution substitute for analysis. In cases like this one, where action and restraint are both risky, the precautionary principle is overly risk-averse and indeterminate: restraint from civil defense could mean investment in other forms of missile defense or in nuclear weapons themselves, which themselves carry escalation risks, as elaborated in Section 5. When precaution is applied only above a certain threshold, the need for numerical judgment is reintroduced without answering which precautions to take, how much to spend, or which portfolio of interventions to choose. This paper attempts to answer those questions.

Expected value calculations, however, cannot be taken at face value when estimates carry high variance. Under a Bayesian framework, priors about how effective certain kinds of interventions tend to be are significantly affected by large variance. Accordingly, alongside the quantitative estimates, we present a robust qualitative case for each intervention, including the mechanisms involved, historical and cross-national analogs, and a defense of the predictions employed. We note that costs tend to be inverse to time: placing a limit on current expenditures carries the incidental benefit of deferring additional expenditures to a time when more efficient solutions are available.⁵⁵ We treat empirical grounding as a constraint on our analysis throughout, explicitly noting where no substantial evidence supports a specific magnitude.

⁵³Carole Bernard, et al., “Catastrophe Aversion and Risk Equity in an Interdependent World,” *Journal of Management Science*, 64, no. 10, 13 Nov 2017; Baruch Fischhoff, et al., “Knowing with certainty: The appropriateness of extreme confidence,” *Journal of Experimental Psychology: Human Perception and Performance*, 3(4), 552–564. doi.org/10.1037/0096-1523.3.4.552

⁵⁴Alasdair Phillips-Robins. “Catastrophic Risk, Uncertainty, and Agency Analysis.” *Institute for Law and AI*. Sept. 2022. law-ai.org/catastrophic-risk-uncertainty-and-agency-analysis/

⁵⁵Richard A. Posner. “Efficient Responses to Catastrophic Risk.” 6. *Chicago Journal of International Law*. 511. 2006. chicagounbound.uchicago.edu/cgi/viewcontent.cgi?article=2895&context=journal_articles

4.3 Probability Inputs

The framework requires a baseline range for the annual probability of a large-scale nuclear attack on U.S. territory, and we rely on the estimates of scholars and forecasters with strong records in geopolitical analysis. The closest available approximation to a meta-estimate is a 2018 Good Judgment survey on the probability of a state-actor nuclear detonation causing at least one fatality, which yielded 0.004 per year.⁵⁶ The Existential Risk Persuasion Tournament asked superforecasters and domain experts the likelihood that nuclear weapons would kill more than 10% of humanity by 2100; the median superforecaster estimate of 4 percent and median expert estimate of 8 percent annualize to roughly 0.0005 and 0.001 per year, respectively.⁵⁷ We extend the range in both directions and adopt a working band of 0.0002 to 0.003 per year. Consistent with the standard defined above, Section 6.2 biases within this band toward the estimates least favorable to further expenditure on alternative civil defense measures.

4.4 Evaluation Framework

We aim to present a robust decision-making strategy by presenting a large ensemble of possible futures, each with an internally consistent set of assumptions, against which a candidate strategy should perform reasonably well while avoiding catastrophic failure in most of them. We also present an adaptable framework by specifying near-term actions and including signposts or triggers, characterizing risk by estimating the maximum likelihood policymakers would have to assign to a strategy’s breaking scenarios before they would rationally choose some other action. The strategy is evaluated under the XLRM framework: X represents exogenous uncertainties, L describes policy levers, R includes relationships such as escalation dynamics and command and control, and M denotes the relevant measures, including survival, recovery, and deterrence stability.⁵⁸ The ultimate product of this methodology is a portfolio of interventions, developed in Section 6 and prioritized in Section 7, that performs acceptably across the ensemble of futures.

5 Why Traditional CD Has Negative EV Under Modern Conditions

As previously established, the expected value of an intervention has two movable terms: the damage it averts and the probability it alters. This section evaluates the traditional Cold War portfolio consisting of mass evacuation and mass sheltering protocols, and public education campaigns, against both terms under modern conditions. This analysis finds that each measure fails on the first term while several actively worsen the second.

⁵⁶Luisa Rodriguez. “How many people would be killed as a direct result of a US-Russia nuclear exchange?” *Rethink Priorities*. June 30, 2019. rethinkpriorities.org/research-area/how-many-people-would-be-killed-as-a-direct-result-of-a-us-russia-nuclear-exchange/

⁵⁷Ezra Karger et al., *Forecasting Existential Risks: Evidence from a Long-Run Forecasting Tournament*, FRI Working Paper #1 (Forecasting Research Institute, July 2023), forecastingresearch.org/research/existential-risk-persuasion-tournament.

⁵⁸Robert J. Lempert, et al. “Shaping the Next One Hundred Years: New Methods for Quantitative, Long-Term Policy Analysis.” RAND Corporation. (Report) 2003.

5.1 Mass Evacuation

Mass evacuation became the Federal Civil Defense Administration's primary program under director Val Peterson, and the reasons were as much fiscal as strategic. Peterson shifted the agency's emphasis away from shelters toward evacuation in the early 1950s, arguing in the 1954 budget proposal that the growing yield of nuclear weapons threatened to turn public shelters in large cities into death traps. The move pleased both the administration and its congressional critics for a simpler reason: evacuation was cheap, and shelters were not.⁵⁹ The policy was proven "dangerously inadequate" within the year, by the 1954 Bravo hydrogen bomb test at Bikini Atoll. Bravo is best known for the Lucky Dragon incident, in which twenty-three Japanese fishermen were contaminated while sailing through the test area, but its more consequential result for civil defense planning was the unexpectedly high level of radioactive fallout produced at ranges approaching one hundred miles from the detonation site. During 1955 congressional hearings, Atomic Energy Commission officials acknowledged that the explosion had generated fallout across a seven-thousand-square-mile area — a range no critical mass of evacuees could clear within any reasonable timeframe.⁶⁰

The objection raised by Peterson's contemporaries was practical as much as physical, and it survives the intervening seventy years intact. Evacuation is slower than sheltering by construction of the latter. Specifically, evacuation requires a large proportion of the population traversing a great distance over a small number of viable routes. Covering one hundred miles by road requires between three and six hours under normal traffic conditions which would not plausibly hold in the wake of a nuclear altercation. Civil defense research shifted back toward a mixture of sheltering and evacuation, with emphasis on the former, after Peterson's resignation. The field, in other words, reached this conclusion contemporaneously and then lost it.⁶¹

The modern warning environment also converts this practical difficulty into an impossibility. If mass mobilization with any hope of success must begin hours, and more likely days, before an anticipated strike, and if ICBMs reach targets thousands of miles away with ten to fifteen minutes of detectable warning, then an effective evacuation must begin before a strike is imminent — which is to say, it is available only to a state that intends to strike first. Yet the mass mobilization of a civilian population would immediately signal hostile intent, undermining the very surprise on which the first strike depends. Evacuation is therefore doubly defeated: it cannot protect a population under modern warning times, and the only scenario in which its timeline is achievable is the scenario in which it functions as a war signal rather than a safety measure.⁶²

5.2 Mass Sheltering

Mass sheltering suffers from a related logistical constraint, though its binding limitation is behavioral rather than geographic. The familial reunification threshold identified by Mawson and discussed in Section 2.3 is unlikely to be met by a critical mass of the population within a fifteen-minute window from warning to blast. Even with designated rendezvous points and known shelter locations, most families cannot reasonably be expected to locate one another and reach a protective structure in that interval, which substantially reduces the efficacy of any

⁵⁹Rose, "One Nation Underground," 2001, 33–35

⁶⁰Ibid. Rose 2001, 33–35.

⁶¹Ibid. Rose 2001, 33–35.

⁶²Schelling, *Arms and Influence*, 240–244.

sheltering strategy predicated on rapid compliance.⁶³

Sheltering is also expensive, which was recognized as the program’s binding political constraint from the beginning. Kahn’s own recommendation, that civil defense expenditure approach five hundred million dollars in 1960 dollars, amounting to roughly \$5.5 billion inflation-adjusted terms — and the difficulty he anticipated was not public opposition to sheltering but public unwillingness to finance it. Garret’s compilation of public attitude studies bears this out: though favorability toward sheltering initiatives ran between sixty and seventy percent at the program’s height, the issue of public financing was consistently its most contested element.⁶⁴

Contemporary cost data confirm the scale of the problem. Swiss figures, drawn from a program that aims to provide a shelter space for the entire national population, indicate construction costs ranging from approximately \$1,000 per space for larger shelters with capacities of fifty to two hundred, up to \$3,000 per space for smaller shelters that capture no returns to scale. Renovating existing basements, mines, and underground transit systems would cost less, but implementing any comparable program at United States scale would still run into the hundreds of billions.⁶⁵

Sheltering carries an additional liability on the probability side of the ledger. As the Soviet case discussed in Section 3.6 demonstrates, shelter-in-place capacity is precisely the form of preparedness that complements a counterforce first strike, because it can be activated quickly and quietly. A national shelter network is thus the variant of traditional civil defense most legible to an adversary as first-strike preparation, a perception the resilience literature independently flags as escalatory (Section 2.5).

5.3 Public Education Initiatives

Public education campaigns, the cheapest arm of the traditional portfolio, fail on more nuanced grounds. On the damage term, their protective value under compressed timelines approaches nil. The behavioral literature reviewed in Section 2.3 documents a persistent knowledge-action gap in which favorable attitudes, and even accurate knowledge, do not convert into effective protective behavior: respondents who identified fallout as a major danger frequently reported that they did not know what to do about it or believed nothing could be done. A fifteen-minute warning window leaves no room for the deliberate recall and execution of trained procedures by an unpracticed public.⁶⁶

The Hawaii false alert of January 2018 supplies the closest available natural experiment, and the result is unfavorable. Of respondents surveyed afterward, only twenty-eight percent initially believed the alert; twenty-seven percent disbelieved it, and forty-five percent were unsure — and many were uncertain about how to act even had they believed it. Mileti and Sorensen anticipate

⁶³Anthony R. Mawson, *Understanding Mass Panic and Other Collective Responses to Threat and Disaster*, 2005.

⁶⁴Herman Kahn 1960, 592; Garret, *Overview of Public Attitude Studies Pertaining to Civil Defense*, 1979.

⁶⁵Switzerland’s program guarantees a shelter place for the entire resident population: roughly 370,000 public and private shelters provide space for about nine million people, a value exceeding the country’s population. Federal Office for Civil Protection, *Shelters: Their Purpose, Construction and Utilisation* (Bern: FOCP, June 2023). On the program’s scale and financing, including the roughly \$900 owed per resident’s spot in a public shelter by those who do not build their own and a \$250 million federal “upgrade,” see Reuters, “Switzerland Plans Revamp of Cold War-Era Nuclear Bunker Network,” December 12, 2024. The scale claim then follows multiplicatively: at any per-place cost in this range, provisioning on the order of 350 million places implies expenditure in the hundreds of billions of dollars, before differences in land, siting, and retrofitting conditions are even considered.

⁶⁶Garret 1979; Murthy 2018.

the structural reason: an impending event may exceed anything historically experienced or planned for, which makes it difficult for warnings to encourage appropriate protective action or to offer precise recommendations at all. Education that has not established credibility in advance does not produce compliance when it is actually needed.⁶⁷

There is an additional point regarding what Cold War programming truly sought to achieve. Government messaging paired an insistence on the immediacy of the threat with an assurance of survivability, on the reasoning that public participation required the risk to be both tangible and immediate. Education of this kind was popular in part because it was compatible with prevailing strategic doctrine rather than anything to do with efficacy. Bert the Turtle’s duck-and-cover belongs to a class of measures that make people feel marginally safer without materially changing their survival prospects — an illusion of safety deployed to manage panic rather than to reduce casualties.⁶⁸

On the probability term, education campaigns are the most visible and most explicitly fear-mediated instrument of civil defense. They operate by making nuclear war imaginable to the public at scale, which is precisely the conceptualizing channel through which preparedness can erode the normative prohibition on use, as discussed in Section 3.4. Furthermore, as a broadcast signal of national preparation, they are legible to adversaries in a way that engages the visibility condition discussed in Section 3.7. Though it is technically possible, as Section 3.5 argues, for preparedness embedded in ordinary civic life to strengthen the taboo rather than weaken it, crisis-framed campaigns popular during the Cold War rest on the wrong side of that distinction. Hence, we have a measure that saves few lives, costs little, and moves the probability term in the wrong direction. This is no bargain but instead a small purchase of negative expected value.

5.4 Framework Results

Evaluated against the conditional factors specified in Section 3.5, the traditional Cold War civil defense portfolio fails comprehensively. Most traditional civil defense initiatives pose a signaling problem, as shelter networks and educational campaigns are public by nature, and evacuations are highly visible to adversaries and could signal preparation for attack. These mechanisms are crisis-deployable, or rather they aspire to be, which is exactly the property that couples preparedness with first-strike calculation. What they protect, urban populations in place, sits at the center of the adversary’s targeting calculus rather than outside it. And their engagement with the taboo runs through crisis-framed fear rather than through the everyday civic embedding that strengthens the prohibition on use. On the damage side of the expected value calculation, modern warning times reduce evacuation to impossibility, reduce sheltering to a reunification problem the behavioral literature says cannot be solved in fifteen minutes, and reduce education to comfort.

The historical judgment, rendered through defunding and the gradual absorption of civil defense into a broader all-hazards approach through FEMA (Section 2.1), was reached on grounds of cost and feasibility alone. The framework developed here demonstrates that the verdict was correct twice over, because the same measures also carried adverse probability effects that the Cold War debate suspected but never systematized. The traditional civil defense portfolio under modern conditions has negative expected value under both terms at once. Hence, we must consider what a reconceptualized modern preparedness portfolio could look like, yielding an evolution within the definition of civil defense.

⁶⁷Ling 2021; Murthy 2018; Mileti and Sorensen, “Communication of Emergency Public Warnings,” 1990.

⁶⁸“Survival Under Atomic Attack,” 1950; Sincere 1984

6 Interventions and Cost Benefit Analysis

As previously noted, civil defense is conceptually associated with the Cold War, and with all the absurd measures that manufactured the illusion of safety, though the foundations were resilience and preparedness. Considering this, we argue that a reconceptualized civil defense portfolio is warranted in modern conditions. Specifically, one that moves beyond the narrow emphasis on evacuation plans, educational programs, and shelters toward a broader portfolio of interventions evaluated on the marginal reduction in expected fatalities per dollar spent, and on their potential to provide ancillary benefits in the case that a large-scale nuclear exchange does not occur. In addition to offering rapid protective actions under compressed timelines to reduce the significant risks associated with short-term blast and radiation exposure, such a program ideally addresses the longer-term, right-of-boom risks associated with the large-scale disruption of critical supply chain networks and institutional stability that would follow from a nuclear attack. Rather than solely emphasizing a reduction in immediate deaths from the initial blast, a robust civil defense program would also focus on mitigating downstream second-order risks by preserving basic societal functions and institutions.

By extending the policy horizon beyond immediate casualties to include longer-term survival and recovery trajectories, such an approach is likely to yield substantially higher expected value of lives saved over time while simultaneously strengthening resilience to a wider class of shocks that could otherwise culminate in famine, disease, or societal collapse. A brief overview of the relevant policy measures constituting such a reconceptualized initiative follows.

6.1 Supply Stockpiling

Maintaining an adequate supply of resources that would become critical has served as a key facet of disaster planning and would also lie at the forefront of a revitalized civil defense initiative. Stockpiling would include, but is not limited to, pharmaceutical countermeasures such as potassium iodide (KI), which blocks thyroid absorption of radioactive iodine-131, and Prussian blue (ferric hexacyanoferrate), which prevents ingestion of cesium-137 by binding the material in the intestines. Widespread distribution of radiation dosimeters shared by small localities, by virtue of allowing individuals and authorities to monitor cumulative exposure levels in real time, could also result in more informed decisions regarding continued sheltering versus re-initiating productive economic behavior in some capacity. Providing these medical and monitoring capacities in afflicted areas during the immediate aftermath of a nuclear strike would not merely marginally improve outcomes but plausibly reduce radiation-induced morbidity and mortality by at least 20%.⁶⁹ Maintaining food supplies is another crucial aspect of this subset of the initiative in light of the climatic effects and likely breakdown of the food supply chain that would follow from a medium to large scale nuclear altercation. Fortunately for the body of research on this topic,

⁶⁹This 20% figure is a conservative synthesis of three independent effect channels. Intensive supportive care roughly doubles the survivable whole-body dose ($LD_{50/60}$ rising from approximately 3.5–4 Gy to 6–7 Gy): Jamie K. Waselenko et al., “Medical Management of the Acute Radiation Syndrome: Recommendations of the Strategic National Stockpile Radiation Working Group,” *Annals of Internal Medicine* 140, no. 12 (2004): 1037–1051. Informed sheltering and shelter-exit guidance (the function of the dosimetry and GIS mapping measures) avoids the majority of potential fallout casualties in modeled urban detonations: Brooke R. Buddemeier and Michael B. Dillon, *Key Response Planning Factors for the Aftermath of Nuclear Terrorism*, LLNL-TR-410067 (Livermore, CA: Lawrence Livermore National Laboratory, 2009). On timely iodine thyroid blocking, see World Health Organization, *Iodine Thyroid Blocking: Guidelines for Use in Planning for and Responding to Radiological and Nuclear Emergencies* (Geneva: WHO, 2017); on radiocesium decorporation, see Rump et al., “Benefit-Cost Analysis of Radiocesium Decorporation.”

food storage has garnered significant attention from policymakers and researchers as the primary agricultural disaster response policy, and also from survivalists and preppers seeking to persist in the case of an emergency. However, such preexisting emphasis suggests that expanding these storage capabilities does not offer a particularly cost-effective alternative relative to other means of food provisioning that will be discussed below. The literature suggests that humanity would currently enjoy at least seven months of adequate food supplies from grain storage and four months from other sources if agricultural productivity were halted.⁷⁰ Denkenberger’s assessment is that fully equitable distribution of existing stocks would sustain the global population for roughly a year, after which supplies would be exhausted entirely, indicating that the benefits of stockpiling interventions are realized almost exclusively in the short term.⁷¹ Hence, further emphasis on longer-term agricultural solutions is necessary.

6.2 Agricultural Resilience

Maintaining some level of U.S. agricultural productivity in the event of a nuclear conflict of any scale presents perhaps the largest reduction in the risk to the long-term survival of postwar civilization by preventing the prospect of mass starvation. Two primary shocks resulting from nuclear war would seriously impact the reliable environmental conditions currently necessary for the functioning of the global agricultural system. First, an abrupt sunlight reduction scenario (ASRS) resulting from the sudden injection of sulfates or soot into the atmosphere, where they would remain for decades: the ensuing reduction in sunlight, temperature, and precipitation under escalated scenarios would reduce the expected per capita caloric intake of most nations below what is necessary to sustain even resting energy expenditure for the current population level.

Second, global catastrophic infrastructure loss (GCIL) could lead to a collapse in agricultural production in light of the current system’s reliance on technical industry in the form of processing machines, vehicles, fuel, fertilizer, and pesticides. Per a 2024 estimation by Moersdorf et al., the absence of synthetic fertilizer, pesticides, irrigation, and mechanization would reduce yields by proportions as large as 35% for rice, 41% for soybeans, 48% for corn, and 37% for wheat.⁷²

Resilient food solutions offer perhaps the most cost-effective measure addressing the risk to human life associated with a large-scale interstate nuclear conflict. These may be defined as interventions capable of producing substantial quantities of calories and essential nutrients when conventional food production methods would be inadequate to feed the global population.⁷³ Methods must meet three thresholds to be classified as a resilient food technology: the method provides sufficient food to feed a significant portion of the global population; consistent

⁷⁰David C. Denkenberger and Joshua M. Pearce, *Feeding Everyone No Matter What: Managing Food Security after Global Catastrophe* (Waltham, MA: Academic Press, 2015), which estimates roughly seven months of consumption from grain storage, drawing on T. Do, Kim Anderson, and B. Wade Brorsen, “The World’s Wheat Supply,” Oklahoma Cooperative Extension Service, 2010 — plus roughly four months from other sources, including household and retail stocks and animal sources. An independent estimate from FAO data places total world food stocks near eight months of consumption: Francesco Laio, Luca Ridolfi, and Paolo D’Odorico, “The Past and Future of Food Stocks,” *Environmental Research Letters* 11, no. 3 (2016): 035010.

⁷¹David C. Denkenberger and Joshua M. Pearce, “Cost-effectiveness of Interventions for Alternate Food in the United States to Address Agricultural Catastrophes,” *International Journal of Disaster Risk Reduction* 27 (2018): 278–89, doi.org/10.1016/j.ijdrr.2017.10.014.

⁷²Jessica Moersdorf et al., “The Fragile State of Industrial Agriculture: Estimating Crop Yield Reductions in a Global Catastrophic Infrastructure Loss Scenario,” *Global Challenges* 8, no. 1 (2024): 2300206, doi.org/10.1002/gch2.202300206.

⁷³Denkenberger and Pearce, “Cost-effectiveness of Interventions for Alternate Food,” 280.

production levels can be maintained amid severe resource constraints; and, most crucially, the method is scalable, meaning a drastic and rapid ramp-up of production, before existing food storage runs out, is a viable alternative.⁷⁴

Further details regarding specific existing methods are offered in the cost-benefit analysis below. It is worth noting that the alternative food production techniques expanded under such a program range from measures as basic as reallocating food currently used as animal feed toward human consumption, to converting leaves and cellulosic sugar, both derived from non-food biomass, into edible products and nutritional ingredients through processing technologies of varying complexity.

6.3 Trade Resilience

Partially redirecting trade policy to account for a potential shock to domestic agricultural production would offer a crucial supplement to the set of policies expanding the scope and production of resilient foods. Because no countries in the Southern Hemisphere currently possess nuclear weapons, and because more land sits closer to the equator and more ocean (which acts as a temperature buffer) lies in the Southern Hemisphere, those countries' agricultural productivity would suffer far less from an ASRS resulting from a nuclear conflict; simulations of yield declines for relevant crops rarely fall beyond the range of 10 to 20 percent for countries below the equator.⁷⁵

Furthermore, per Rosenberger, writing in 2024, U.S. vulnerability to an ASRS is disproportionately associated, relative to other countries, with the projected loss of critical imports. Such results suggest that the U.S. would enjoy key benefits from enhancing trade relationships to capture a higher percentage of the remaining exportable surplus of foodstuffs from Brazil, Argentina, and Australia, all of which would retain a relatively higher degree of productive capacity during an ASRS.⁷⁶

6.4 Enhancing Post-Detonation Response Capacity

The final facet of a reconceptualized portfolio is neither material nor infrastructural but informational and organizational: the capacity of the population and its institutions to receive, believe, and act on accurate information in the hours and weeks following a detonation. The behavioral

⁷⁴Juan B. García Martínez, Jeffray Behr, Joshua Pearce, and David Denkenberger, "Resilient Foods for Preventing Global Famine: A Review of Food Supply Interventions for Global Catastrophic Food Shocks Including Nuclear Winter and Infrastructure Collapse," *Critical Reviews in Food Science and Nutrition* (published online February 11, 2025), doi.org/10.1080/10408398.2024.2431207.

⁷⁵Jonas Jägermeyr et al., "A Regional Nuclear Conflict Would Compromise Global Food Security," *Proceedings of the National Academy of Sciences* 117, no. 13 (2020): 7071–7081; Lili Xia et al., "Global Food Insecurity and Famine from Reduced Crop, Marine Fishery and Livestock Production due to Climate Disruption from Nuclear War Soot Injection," *Nature Food* 3 (2022): 586–596. The 10–20 percent range characterizes Southern Hemisphere production under small and moderate soot injections; under the largest plausible injection scenario of 150 Tg, absolute declines are larger everywhere, though the Southern Hemisphere's relative advantage persists.

⁷⁶James Rosenberger, "Measuring Community Significance and State Vulnerability after Nuclear Winter" (working paper, Existential Risk Laboratory, The University of Chicago, August 2024). In Rosenberger's vulnerability decomposition the United States is distinctive in that ranked import declines contribute more to its score than domestic yield declines (55.69 percent of the U.S. score in the 5 Tg case; 43.80 percent at 47 Tg, led in both by wheat), unlike nearly all other profiled states. His analysis builds on the post-catastrophe yield data of Xia et al., "Global Food Insecurity and Famine," and the trade-network framework of Felix U. Jehn et al., "Food Trade Disruption after Global Catastrophes" (preprint, EarthArXiv, 2024). The retained productive capacity of Brazil, Argentina, and Australia under an ASRS is established Xia et al., "Global Food Insecurity and Famine."

literature reviewed in Section 2.3 identifies warning credibility as the decisive bottleneck of any protective response, and the 2018 Hawaii false alert demonstrated that a technically functional warning system commands little belief in the absence of a preparedness framework that makes protective action seem possible.

Modern conditions favor this intervention class in a way they disfavor the traditional portfolio: alert systems are now smartphone-based rather than siren-based, meaning information can be dispersed more rapidly, specifically, and effectively than in any previous era, and much existing infrastructure is dual-use, with underground metro systems such as Washington D.C.’s capable of serving as fallout shelter despite not being designed for it. Response capacity therefore comprises three components. First, warning-system credibility: sustained investment in the accuracy, specificity, and drill-tested familiarity of wireless emergency alerts, so that a real warning is given credence. Second, shelter and resource mapping: a national Geographic Information System project providing an accessible, interactive map of viable shelter and, post-strike, of resource locations, enabling the nonlethal movement of survivors toward supplies as an alternative to starvation in place. Third, post-strike coordination protocols, including guidance on safe shelter-exit timing, which couples well with the dosimeter distribution described above.

The emphasis on organization over construction follows the resilience literature’s central finding that no amount of material preparation functions without maintenance, training, and institutional knowledge, while organizations that train for disruption can improvise even when materials are lacking, as discussed in Section 2.5. Critically, this intervention class is favorable under the conditional factors of Section 3.7, as its components are domestically visible but strategically illegible. That is, credible alerts and shelter maps can neither protect strike capability nor drastically expand usage as a prelude to attack, and their everyday civic embedding engages the taboo-strengthening channel rather than the crisis-fear channel that traditional education campaigns activated.

6.5 Cost-Benefit Analysis

The analysis proceeds on two tracks that answer two different questions. The first track estimates the cost of each intervention in the reconceptualized portfolio individually, producing the cost-per-intervention figures from which Section 7 derives its cost-effectiveness ranking. The second track, following the standard defined in Section 4.1, evaluates program-level packages: it computes the expected annual loss under three escalating levels of national preparedness and derives, for each, the maximum increase in the annual probability of nuclear attack that the program could impose before its expected losses exceed those of minimal response. This second track is the operational form of the acceptable-risk question, and its bounds are compared directly against the probability shifts plausibly attributable to civil defense through the mechanisms of Section 3.

Where an intervention appears on both tracks it is costed at different configurations by design: community dosimeter coverage, at one device per geographic cluster of one thousand people, is the minimal configuration costed individually below, while the program-level cases price denser individual coverage at one device per four people as part of a threshold package. Likewise, the standalone GIS mapping estimate below covers the mapping initiative alone, while Case 2 prices a larger bundle of mapping, signage, and alert-system integration. These are different purchases, not conflicting estimates of the same one.

6.5.1 Intervention-Level Costs

Consideration of the relative costs and benefits of implementing certain policies considering a nuclear conflict can offer critical evidence for and against certain civil defense investments. Such analysis is necessarily imperfect; we cannot “play God” and weigh every possible scenario disseminating from a minute policy change. However, even under conservative assumptions regarding the annual probability of a nuclear detonation and the effectiveness of interventions, this expected value framework suggests that the relatively low-cost measures which reduce immediate blast and radiation exposure, sustain food production in light of an abrupt reduction in sunlight or severe reduction in infrastructure, or otherwise accelerate the nation’s recovery can yield extraordinarily large, expected benefits. That is, such analysis may demonstrate the economic viability of these policies considering the magnitude of the risks involved and help ensure that public resources are allocated in an efficient manner. We proceed in order of the policies presented under the reconceptualized notion of civil defense.

First, regarding emergency pharmaceuticals, current prices for standard packs of non-perishable potassium iodide tablets suggest an average cost of \$1 to \$1.50 per 130 mg tablet, which may be consumed on a single-use basis shortly after initial exposure to iodine-131. Hence, targeted coverage for the ~50 million survivors of the initial blast most plausibly exposed to meaningful radioactive iodine risk would cost closer to \$75 million when proceeding with the upper-bound estimate to account for storage costs. As shown by Rump et al., Prussian blue treatment is a far more intensive and costly process: rather than being single-use, the treatment process, which ideally begins directly after exposure, lasts a minimum of 30 days, with a recommended daily consumption of 9 grams; with the average 18-gram bottle costing roughly \$150, the total treatment cost per person is unlikely to be lower than \$2,250.⁷⁷ Applying the same linear scaling principle as previously suggested a total cost of ~\$115 billion for a robust Prussian blue stockpiling initiative.⁷⁸

Public knowledge of safe times to exit shelter and resume ordinary activity would also serve as a crucial step toward recuperation among those who survived the initial blast. The implementation of community dosimeters would likely serve as the most cost-effective measure ensuring this possibility. Dosimeters of decent quality typically cost between \$150 and \$300, but the group of survivors in any given community needs only one, assuming communication between survivors in different shelters remains relatively strong in light of the mapping proposal introduced below. After several days of a prolonged crisis, a rotation of one person checking radiation levels outside every several hours could reasonably begin: per acute radiation syndrome (ARS) thresholds, exposure below 1 Gray (Gy) will not exceed mild ARS, which is not life-threatening in the short term, and as fallout levels fall by a factor of ten for every sevenfold increase in hours after the blast, the radiation level after one week is under 0.1% of initial levels. Typical dose rates in heavy fallout zones fall from 1–10 Gy/hour an hour after the blast to roughly 0.01–0.1 Gy/hour after one week, suggesting that open-air movement in even a relatively high residual hotspot for up to an hour would not result in anything beyond an extremely minor degree of ARS. We therefore suggest that one dosimeter per geographically clustered group of 1,000 people is sufficient, meaning a comprehensive program would have an expected cost range of \$52,000,000 to \$105,000,000.

Digital shelter mapping could also serve as a cost-effective measure leading to a significant

⁷⁷A. Rump, et al. Benefit-Cost Analysis of Radiocesium Decorporation by a Prussian Blue Treatment and Stockpiling. *Journal of Drug Research*, 68(2), 89–99, 2018. doi.org/10.1055/s-0043-119793.

⁷⁸Ibid. Rump, et al., 2018.

relative increase in survival among those who made it past the initial blast but face severe resource shortages within days. Supposing that survivors had some means of communication with one another, movement toward public shelters with more ample resources days after the initial blast could serve as a worthwhile alternative. This prospect of nonlethal movement alludes to the potential benefits of a national Geographic Information System (GIS) project providing an accessible, interactive map of viable shelters for survivors with dwindling supplies. Grant sizes and cost estimates on planning documents suggest that regional GIS projects typically cost on the order of \$1 million to \$10 million for state projects associated with the Department of Transportation. Multiplying a relatively low per-state estimate of \$5 million (assuming benefits from scale) by fifty and adding costs associated with federal integration and maintenance yields a total estimated cost of \$300 million for such a mapping initiative.

Lastly among the stockpiling measures, the cost-effectiveness of adequate food storage wanes in comparison to virtually all other alternatives. Though prices of stored food can be reasonable, with the Latter-day Saints (LDS) Church maintaining an online store featuring a case of 28 lbs. of pre-packaged product for \$33.50, the cost of private storage is prohibitive for most Americans. As noted by Denkenberger et al., adequate supplies for a family of four to subsist for five years would require an upfront investment of more than \$12,000 on a quantity of stored food sufficient to fill a bedroom, and even these extremely well-prepared families would likely starve after supplies ran out under more severe nuclear scenarios absent robust alternative food production.⁷⁹ Furthermore, large-scale storage measures could exacerbate present food insecurity by creating an artificial scarcity that would increase the price of food. It is worth noting that some degree of food storage may provide meaningful resilience to shorter-term supply-chain disruptions and less extreme catastrophes: at the individual level, a modest quantity of stored food offers a critical time buffer for the deployment of more scalable technical solutions. However, food, by virtue of being more voluminous and perishable than the other goods, is significantly more costly to store and distribute at scale, limiting its viability as a key part of a government-led preparedness strategy. While some reserves may be justified on the grounds of their capacity to address short-term disruptions, large-scale public stockpiling is unlikely to serve as an effective primary pillar of civil defense relative to interventions that revitalize agricultural production or reduce mortality directly.

The agricultural resilience interventions introduced in Section 6.1.2 present a cost profile unlike any other measure in the portfolio, because what is purchased up front is not capital stock but readiness: research, development, and response planning for the rapid scale-up of resilient food production. Cost-effectiveness analyses of this intervention class estimate that a preparedness investment on the order of \$100 million, covering research and development on alternative food pathways (e.g., agricultural residue and leaf protein extraction, cellulosic sugar conversion, methane single-cell protein, and seaweed cultivation among them), response planning, and integration with existing agencies, would meaningfully raise the probability that scalable production is achieved before stored food runs out.⁸⁰ Monte Carlo modeling of the U.S. case places the expected cost per life saved in the present generation between \$1 and \$20,000 even at the pessimistic bound.⁸¹ Set against the \$11.7 million value of a statistical life used throughout this analysis, the ratio is not close: resilient-food preparedness is, by two to seven orders of magnitude, the most cost-effective intervention in the reconceptualized portfolio, and its expenditure

⁷⁹Denkenberger and Pearce, “Cost-effectiveness of Interventions for Alternate Food,” 280.

⁸⁰Ibid. Denkenberger and Pearce, 280.

⁸¹Joshua M. Pearce & David C. Denkenberger. “A National Pragmatic Safety Limit for Nuclear Weapon Quantities.” *Safety* 2018, 4(2), 25; doi.org/10.3390/safety4020025.

profile, consisting of laboratory research, pilot facilities, and planning documents, is invisible to adversary targeting calculus in the sense defined in Section 3.5. Consistent with our practice of biasing against our own conclusions, we note the principal uncertainty: these estimates assume that preparedness translates into deployment during a catastrophe, a translation that depends on the institutional continuity that the response-capacity interventions above exist to preserve. The portfolio’s components are, in this respect, complements rather than substitutes.

Though a doctrinal shift in trade would not come with any direct costs, ensuring reliable food imports under conditions as extreme as a nuclear scenario would very likely require pre-negotiated agreements with exporters among key Southern Hemisphere producers such as Brazil, Argentina, and Australia, granting the U.S. the right to procure large quantities of food on short notice. Such contracts would ideally include priority clauses ensuring U.S.-bound shipments are fulfilled ahead of other buyers, and a price roof to limit exposure to a likely spike. The counterparties must receive compensation for holding inventory or reserving export capacity in light of a dramatic price increase, and for bearing significant political risk in potentially facing an export ban in the wake of a disaster. Annual cost estimates for each country therefore emerge from assigning an option premium of 20% to the present market value of an import quantity that would satisfy $\sim 35\%$ of national caloric needs, corresponding to roughly 25 million tons of imported food goods: 15 million tons from Brazil at a present market value of $\sim \$5$ billion (premium $\sim \$1$ billion), 5 million from Argentina at $\sim \$2$ billion (premium $\sim \$400$ million), and 5 million from Australia at $\sim \$3$ billion, accounting for higher market and transportation costs (premium $\sim \$600$ million). As a result, this research suggests that pursuing an extensive trade relationship with Brazil, Argentina, and Australia featuring robust disaster terms would carry an additional annual cost on the order of $\$2$ billion.

6.5.2 Program-Level Expected Loss and Acceptable-Risk Bound

We now compute, per the standard of Section 4.1, the maximum acceptable increase in the probability of nuclear crisis under escalating levels of national preparedness, seeking in each case the point at which the expected losses of a civil defense initiative exceed those of minimal response. Every case is evaluated with the same loss function:

$$L = C + p \cdot D_{\text{dam}}, \quad (32)$$

where C is certain yearly expenditure (one-time costs amortized over ten years, plus annual costs), p is the annual probability of the reference attack, and D_{dam} is the damage if it occurs: deaths priced at $\$11.7$ million, closer towards to lower bound for the Value of a Statistical Life (VSL) used by U.S. government agencies,⁸² plus infrastructure repair. The reference attack, unless a case states otherwise, features 150 cities and 400 U.S. military bases struck with 30 minutes of warning, against a population of 350 million. Damage terms account direct mortality and infrastructure repair. Second-order mortality from famine and institutional collapse, the

⁸²The figure is the U.S. Environmental Protection Agency’s central estimate — a default of $\$7.4$ million in 2006 dollars, adjusted for inflation and real income growth to approximately $\$11.7$ million — which sits below the central values of the other principal regulatory agencies: HHS’s current guidance uses $\$13.1$ million in 2023 dollars, and DOT’s updated guidance value is higher still. See U.S. Environmental Protection Agency, “Mortality Risk Valuation,” epa.gov/environmental-economics/mortality-risk-valuation; U.S. Department of Health and Human Services, *HHS Standard Values for Regulatory Analysis, 2024* (Washington, DC: Office of the Assistant Secretary for Planning and Evaluation, January 2024); U.S. Department of Transportation, *Departmental Guidance: Treatment of the Value of Preventing Fatalities and Injuries in Preparing Economic Analyses* (Washington, DC: March 2021).

margin on which the food-system interventions act, is treated separately at the end of this subsection. Consistent with the standard of Section 4.1, wherever a range exists we bias cost and probability estimates toward the strongest possible argument for no further expenditure, including rounding mortality down.

Case 1: No Further Expenditure. Certain expenditure is FEMA’s roughly \$4 billion in general emergency preparedness (FY 2025);⁸³ one-time costs are sunk and excluded. We take the lower bound of Section 4.3, $p = 0.0002$, for the annual probability of large-scale nuclear detonation on U.S. territory. For mortality we draw on Kahn’s “Report on a Study of Non-Military Defense,”⁸⁴ reduced generously for the existence of FEMA: Kahn’s estimate of nearly 90 percent mortality assumes minimal warning, minimal protective behavior, and substantial secondary mortality, whereas the contemporary United States has nationwide alerts, some disaster-response infrastructure, and structures capable of meaningful radiation shielding, and even limited shelter-in-place within a 30-minute window reduces fallout exposure by orders of magnitude. We therefore use 30 percent, roughly corresponding to 100 million people, leading to \$1,170 trillion in mortality damages. Infrastructure adds \$3.95 trillion: 25 percent of urban civilian infrastructure in each attacked city requiring significant repair (\$25 billion per representative city) and \$500 million per base with every counterforce strike assumed successful. Hence $D_{\text{dam}} \approx \$1,174$ trillion and

$$L_1 = \$4\text{B} + 0.0002 \times \$1,174\text{T} \approx \$238.8 \text{ billion per year.} \quad (33)$$

Case 2: Expenditure Past the First Threshold. This program ensures that a significant share of the population can identify and reach adequate shelter within roughly 15 minutes and comply with executive guidance. Crucially, this comes in the form of enhanced awareness and access to existing structures rather than bunker construction. Costs come in the form of \$1 billion one-time for nationwide shelter mapping, signage, and alert-system integration; up to \$500 million per year in maintenance and training; and rudimentary dosimeters at one device per four people, roughly 90 million units featuring a per-unit cost of \$50–70, adding about \$5 billion one-time. Meanwhile, fallout radiation is the primary driver of deaths outside immediate blast zones (15–20 percent based on survey data from Hiroshima and Nagasaki),⁸⁵ so mapping, signage, dosimetry, and a behavioral campaign plausibly cut mortality from 30 percent to 10–15 percent. We use 15 percent — 52.5 million deaths — consistent with substantial but incomplete compliance. Then $D_{\text{dam}} \approx \$618.2$ trillion, and certain yearly cost is $C_2 = \$6\text{B}/10 + \$4.5\text{B} = \$5.1$ billion. The bound follows from $L_2 < L_1$, implying that:

$$p_2 \leq \frac{\$238.8\text{B} - \$5.1\text{B}}{\$618.2\text{T}} = 0.000378. \quad (34)$$

⁸³Federal Emergency Management Agency, “Fiscal Year 2025 FEMA Standard Terms and Conditions,” fact sheet, August 4, 2025. [fema.gov/fact-sheet/fiscal-year-2025-fema-standard-terms-and-conditions](https://www.fema.gov/fact-sheet/fiscal-year-2025-fema-standard-terms-and-conditions).

⁸⁴Herman Kahn, “Report on a Study of Non-Military Defense,” *RAND*, July 1958. rand.org/content/dam/rand/pubs/reports/2008/R322.pdf.

⁸⁵United States Strategic Bombing Survey, *The Effects of Atomic Bombs on Hiroshima and Nagasaki* (Washington, DC: U.S. Government Printing Office, 1946), sec. II (“Flash burns, 20 to 30 percent. Other injuries, 50 to 60 percent. Radiation sickness, 15 to 20 percent”). Both bombs were airbursts producing little local fallout, meaning that nearly all of these radiation deaths reflect prompt radiation. The figure is therefore a lower bound on radiation’s share in the scenarios modeled here, as ground bursts are likely to generate substantially greater delayed fallout. On the dependence of fallout on burst height, see Samuel Glasstone and Philip J. Dolan, *The Effects of Nuclear Weapons*, 3rd ed. (Washington, DC: U.S. Department of Defense and U.S. Energy Research and Development Administration, 1977), chap. 9.

It is worth noting that the power-law model of Section 3.3 assigns a probability of 0.000362 to a 52.5-million-death conflict, meaning that even if the program shifted the probability of a war of that magnitude all the way to the level the historical curve assigns such wars, the increase in probability is still “acceptable” under our standards.

Case 3: Expenditure Past the Countervalue Targeting Threshold. This program shelters the entire at-risk population. From Swiss data (Switzerland aims to shelter its whole population), construction runs \$1,000 per spot in shelters of capacity 50–200 and \$3,000 per spot in small shelters without returns to scale.⁸⁶ With roughly 100 million Americans having basement access (80 percent in standalone homes, 42 percent of those with basements), retrofitting 33 million households at the upper end of FEMA safe-room costs is \$165 billion; large-scale shelters for the remaining 250 million add \$400 billion; the mapping, signage, and dosimeter package adds \$6 billion. These constitute a one-time cost of \$571 billion, roughly ten percent of annual federal spending. Annually, we have the \$4 billion FEMA baseline plus an additional \$1 billion in variable expenditure including shelter maintenance. Mortality: we estimate conservatively that 5 percent of the population fails to reach shelter or succumbs to radiation, which corresponds to 17.5 million deaths. The damage term must also price adversary adaptation: an adversary that observes the program shifts emphasis to counterforce, so we assume all 1,400 U.S. military bases worldwide are struck at \$500 million each, alongside 50 cities at \$25 billion each. Then $D_{\text{dam}} \approx \$206.7$ trillion, $C_3 = \$57.1\text{B} + \$5\text{B} = \$62.1$ billion, and

$$p_3 \leq \frac{\$238.8\text{B} - \$62.1\text{B}}{\$206.7\text{T}} = 0.000855, \quad (35)$$

a substantially higher tolerance than the first threshold. Note that p_3 exceeds the power-law probability of a 17-million-death conflict, 0.00066, meaning that the program is again technically “acceptable.”

The cases above evaluate the tentative portfolio’s visible half, and the acceptable-risk bound exists because visibility is what permits an adversary to price a measure into its targeting calculus. The recovery-capacity measures — agricultural resilience and readiness ($\sim \$100\text{M}$), potassium iodide stockpiling ($\sim \$75\text{M}$), community dosimetry ($\$52\text{--}105\text{M}$), and GIS shelter and resource mapping ($\sim \$300\text{M}$), roughly \$600 million one-time, plus Southern Hemisphere trade agreements at roughly \$2 billion per year in option premiums — sit outside this strategic adversarial logic, meaning that no influence over probability will be considered.

By the criteria of Section 3.7, they enter D (namely via reducing it) without entering F : laboratory research, pilot facilities, planning documents, and option contracts neither preserve strike capability nor have use capabilities that can be greatly enhanced in a prelude to attack. Naturally, a measure that cannot raise the probability of war has no acceptable-risk bound to satisfy. The only program-level question left is whether its certain cost exceeds its expected benefit.

That comparison runs as follows. We conservatively assume the benefit is conditional on the attack occurring, so the expected yearly benefit is $p \cdot N \cdot \text{VSL}$, where N is the number of second-order deaths the measure averts given the reference attack. We hold p at the lower bound of

⁸⁶For the Swiss program and per-place cost data, see note 65 above (Federal Office for Civil Protection, *Shelters*; Reuters, “Switzerland Plans Revamp”). Housing figures are from the U.S. Census Bureau’s American Housing Survey (single-family detached share of the housing stock and presence of basements). Residential safe-room construction cost ranges are from Federal Emergency Management Agency, *Taking Shelter from the Storm: Building or Installing a Safe Room for Your Home*, FEMA P-320, 5th ed. (Washington, DC: FEMA, 2021).

0.0002 used throughout, since a lower probability reduces the expected return on preparedness. The package breaks even when expected benefit equals its amortized yearly cost of \$60 million:

$$N = \frac{\$60\text{M}}{0.0002 \times \$11.7\text{M}} \approx 26,000 \text{ deaths averted, conditional on the attack.} \quad (36)$$

The trade premium, meanwhile, recurring at \$2 billion per year, breaks even at $N \approx 855,000$. These thresholds should be read in consideration of exposure deliberately excluded by the cases above. Every damage term only counts direct deaths. However, under the reference attack roughly 245 million people survive into a landscape where fully equitable distribution of existing food stocks feeds the population for about one year (Section 6.1), an ASRS reduces caloric production below resting energy expenditure (Section 6.2), and pre-arranged imports could cover roughly 35 percent of national caloric need (Section 6.3). Tens of millions of preventable second-order deaths are plausible.

If readiness and trade together avert even one percent of this exposure (~ 2.5 million lives saved), then the package clears the breakeven threshold by two orders of magnitude and the trade premium by a factor of three. Two key features pertaining to this estimate are worth noting. First, there is no double-counting against the cases above, because famine mortality never entered their damage terms. Second, the figure is biased downward twice over. It ignores the package’s value against non-nuclear catastrophes, in addition to any potential probability reduction from the taboo-strengthening channel. These considerations influence much of the policy considerations to follow.

7 Findings and Recommendations

This research yields five findings. First, the traditional Cold War portfolio of mass evacuation, mass sheltering, and public education campaigns carries negative expected value under modern conditions, and on both terms of the calculation: compressed warning times reduce its protective value to near nil, while its visibility, crisis-deploy ability, and fear-forward character move the probability of war in the wrong direction (Section 5). Second, a reconceptualized portfolio passes the acceptable-risk standard of Section 4.1 even underestimates biased consistently against it: the probability increases plausibly attributable to preparedness through the mechanisms of Section 3, benchmarked against the power-law model, fall within the bounds derived in Section 6.5.2 for both program levels evaluated. Third, the portfolio’s interventions are radically unequal in cost-effectiveness. Agricultural readiness is the outlier, purchasing expected lives at \$1 to \$20,000 each against the \$11.7 million value of a statistical life used throughout, followed by potassium iodide stockpiling, community dosimetry, and shelter mapping, each in the tens to hundreds of millions of dollars; at the other extreme, Prussian blue stockpiling approaches \$115 billion and universal sheltering exceeds \$565 billion in upfront costs. Fourth, the risk-benefit paradox is real but designable-around: the interventions that perform best on cost per expected life saved are also, without exception, the interventions most inert under the conditional factors of Section 3.7. This is not a coincidence, as measures that protect recovery capacity rather than populations in place are both cheaper per life saved because they address the second-order mortality that dominates severe scenarios, and strategically illegible, as they never enter an adversary’s targeting calculus. Fifth, and least comfortably, several intuitive measures fail the analysis: large-scale food stockpiling is dominated by agricultural readiness and risks creating the scarcity it insures against, and universal sheltering, the emblem of civil defense in the public imagination, is the portfolio’s worst performer once adversary adaptation is priced in.

These findings support a three-tier structure of recommendations. Tier I comprises the measures we recommend for immediate implementation: a federally funded agricultural readiness program covering resilient-food research, development, pilot facilities, and response planning, on the order of \$100 million; potassium iodide stockpiling for the plausibly exposed population, roughly \$75 million; community dosimeter distribution at one device per thousand residents, \$52 to \$105 million; and the national GIS shelter and resource mapping initiative, roughly \$300 million. The full tier costs under \$600 million, is invisible to adversary planning, and carries substantial ancillary value: every component functions identically for pandemics, volcanic winters, and the climate-driven disasters whose rising frequency the country already experiences. Tier I is, in the vocabulary of Section 4, robust: there is no plausible future in which it produces significant regret.

Tier II comprises measures we recommend conditionally, sequenced behind signposts consistent with the adaptive framework of Section 4.4: the first-threshold public preparedness package of Case 2 (shelter identification, signage, alert-system integration, and behavioral guidance, roughly \$1 billion upfront and \$500 million annually), densified individual dosimeter coverage (\$4.5 to \$6 billion), and pre-negotiated Southern Hemisphere trade agreements (approximately \$2 billion annually in option premiums). These measures remain within the acceptable risk bound but are more publicly visible than Tier I, engaging the taboo and perception channels of Section 3 in ways that warrant monitoring rather than assumption. Appropriate signposts for activation include sustained deterioration in the strategic indicators identified in Section 2.4, such as the continued absence of any successor arms control framework, or forecaster consensus shifting the probability band of Section 4.3 toward its upper bound.

Tier III comprises measures we recommend against at present: Prussian blue stockpiling at national scale, universal shelter construction, and large-scale public food stockpiling. Each fails on cost-effectiveness, and universal sheltering fails twice, since it is the variant of preparedness most legible as first-strike preparation and the one for which Case 3 explicitly prices adversary adaptation. We emphasize that Tier III is a verdict about the present margin, not about the interventions in themselves; the signpost structure permits revisiting them, and Case 3's arithmetic shows that even the maximal program is not irrational, merely dominated.

Interventions	Cost	Basis	Visible to adversary planning	Tier
Agricultural Readiness	~\$100M	One-time program	No	I
Potassium iodide stockpiling	~\$75M	One-time	No	I
Community dosimetry (1 per 1,000)	\$52–105M	One-time	No	I
GIS shelter and resource mapping	~\$300M	One-time	No	I
First-threshold preparedness package	~\$1B \$500M/yr	+ One-time + annual	Yes	II
Densified dosimetry (1 per 4)	\$4.5–6B	One-time	Partially	II
Southern Hemisphere trade agreements	~\$2B/yr	Annual option premiums	Partially	II
Prussian blue stockpiling	~\$115B	One-time	No	III
Universal shelter construction	~\$565B	One-time	Yes	III
Large-scale public food stockpiling	Not costed	—	Partially	III

Table 1: The reconceptualized portfolio by cost, visibility, and recommendation tier.

8 The Fate of Resilience

8.1 Questions the Field Stopped Asking

Civil defense is a strange concept to have believed in. Most of the public today could not readily imagine sheltering in their own homes for weeks or ducking beneath a desk while a mushroom cloud erupts in the distance, and fewer still could imagine taking Bert the Turtle seriously. Although it feels like a piece of bizarre American history, civil defense was not a joke to the strategists that conceived of it. In the shadow of the most destructive weapon ever made, a set of unconventional strategists held that the public could survive a nuclear war and eventually flourish long after one, and they argued this position on strategic rather than humanist grounds. Whether these strategists were correct is beside the point, though it is worth considering that the field never determined whether they were right or wrong; civil defense was permanently unsettled. As Section 2 established, the Cold War debate over civil defense ended in de-funding rather than resolution: the program was dismantled on grounds of cost, feasibility, utility, political exhaustion, and the strategic ambiguity it raised—whether damage limitation raises the probability of the damage it limits—was left open and then forgotten.

Many of the propositions the field treats as settled—that post-detonation interventions mean thinking about the unthinkable and that thinking about the unthinkable erodes the nuclear taboo—are inherited assumptions rather than answered questions, and this inheritance has narrowed what the field is able to investigate epistemologically. It is worth being precise about which part of that inheritance this paper disputes, because the two halves are usually stated as one. The taboo itself is real, and this paper treats it as such: it appears in Section 3.5 as a formal damping mechanism, it does work the deterrence literature cannot do without, and the portfolio recommended in Section 7 was designed specifically to leave it intact. What this paper disputes are the inference drawn from it; that studying how deterrence might fail is itself a form of erosion, while this claim has never been properly tested. It has, instead, been institutionalized,

and the clearest evidence of the institutionalization is methodological. If it was a method the field once considered and questioned, the nuclear field today should reconsider it for a modern world and an increasingly complex nuclear risk landscape.

8.2 Failure Modes of Deterrence

Among the concepts the field has quietly lost is the art of considering deterrence failure modes: the question of not whether deterrence might fail but how, through what sequence, and why. The question is not as idle as the field's perception may suggest, as every intervention evaluated in this paper is priced against a particular way that failure unfolds, and it is unusually difficult to study, since the outcome of interest has never occurred. Wargames remain the primary experimental method available to the nuclear field; wargames are the only method by which theories of nuclear risk can be subjected to something resembling a trial, for the obvious reason that the alternative is unavailable (and irresponsible).

Cold war-era wargames focused on observing how decision-makers functioned under deep complexity and uncertainty. In the era of Kahn, Schelling, and Bloomfield's political-military exercises, games were designed to be played through nuclear use, through warfighting, and through the political and strategic conditions of a post-detonation environment; the point of the exercise was to discover what participants would do after the threshold was crossed.⁸⁷ Contemporary practice, however, has narrowed as it has evolved in a changing landscape where the nuclear taboo is deeply ingrained in how we approach nuclear risk. Most modern wargames are designed to model escalation and de-escalation dynamics and terminate at or before the nuclear-use threshold, with explanations often being cultural rather than analytical. Nuclear use is treated as something we need not put players through over the course of the wargame.⁸⁸ Indeed, intuition could very likely lead to the conclusion that engaging with the aftermath of nuclear use is a direct challenge to the efficacy of work in prevention policy. Such a criticism of the civil defense project and other pathways within post-detonation research, broadly falling into the camp colloquially known as the "moral hazard" critique, warrants further consideration.⁸⁹

8.3 The Moral Hazard of Post-Detonation Interventions

As discussed previously, the moral hazard critique of post-detonation and right of boom interventions argues that the act of investing in post-detonation itself creates a structural moral hazard by rewarding failure and underfunding foundational safety.⁹⁰ This research does not argue in favor of abandoning prevention-based risk reduction in favor of preparation. Rather, it seeks to

⁸⁷Sharon Ghamari-Tabrizi. "The Worlds of Herman Kahn: The Intuitive Science of Thermonuclear War." *Harvard University Press*. 2005; Schelling's own gaming work; Herman Kahn, thinking of the unthinkable; Bloomfield and Whaley, 1965; Brody, 1963; Hermann and Hermann, 1967

⁸⁸On contemporary wargaming practice as a method of international relations research, see Erik Lin-Greenberg, Reid B. C. Pauly, and Jacquelyn G. Schneider, "Wargaming for International Relations Research," *European Journal of International Relations* 28, no. 1 (2022): 83–109. The closest direct evidence on nuclear use within games is Reid B. C. Pauly, "Would U.S. Leaders Push the Button? Wargames and the Sources of Nuclear Restraint," *International Security* 43, no. 2 (Fall 2018): 151–192, which finds in the archival record of Cold War-era political-military games that players proved consistently reluctant to authorize nuclear use even when the games permitted it.

⁸⁹Reid Pauly and Erik Lin Greenberg

⁹⁰See the discussion and sources in Section 2.2 above: Baker, "On the Genealogy of Moral Hazard"; Dembe and Boden, "Moral Hazard: A Question of Morality?"; Benson, *Constructing International Security*; Schwartz, *Right of Boom*; Ruhl, "Philanthropy to the Right of Boom."

redefine our idea of civil defense into something that can be complementary to prevention: if prevention-based risk reduction is not working effectively under modern geopolitical conditions, then neglecting the post-detonation landscape becomes itself a risk multiplier.

The frameworks and structures that govern nuclear risk over time can produce blind spots due to norms and values that become ingrained into the field's practices. The field's assumptions and broad conclusions often preclude us from making progress on risk reduction in a changing landscape. The rise of AI as a research and risk area of inquiry has brought with it a divergent conceptualization of nuclear risk and the treatment of catastrophic risks more broadly. AI safety researchers often treat the advent of superintelligence as inevitable, thus, planning for the post-AGI world and thinking of what the landscape will look like *is* necessary. In essence, preparation for the aftermath is seen as responsible, but this does not mean it takes resources and attention away from AI safety and preventing catastrophic AGI outcomes. The same consideration is not held by researchers in the nuclear risk field.

The moral hazard argument, then, may be considered merely an expected product of a longstanding epistemological culture; one that has existed since 1945, and has experienced little structural change despite drastic changes to the nuclear risk landscape. The bipolar world of our Cold War successes no longer exists, and we are moving into an era characterized by failing nonproliferation efforts, and the death of arms control. Is it not the responsible option to, at the very least, consider alternative approaches when the ones we know intimately no longer work, not as a response a failure of decision-making but rather the drastically changing and evermore complex geopolitical environment?

A posture treating prevention's failure as unthinkable is a bet, placed on behalf of everyone, that deterrence, diplomacy, nonproliferation, and arms control will hold indefinitely. Unfortunately, this bet is being renewed at the precise moment the architecture supporting those instruments has lapsed. The extensive history of near misses and close calls suggests that accidents indeed happen, that a state ought to be prepared for the failure of their primary strategy, and that conflating civilizational collapse with extinction (making preparation appear futile), is a profound epistemic error. The absence of nuclear use from wargaming methods illustrates what this moral posture costs in practice: not merely an unfunded portfolio but a field that has arranged its methods so as to not learn the thing it claims to know. Refusing to prepare for such scenarios only preserves an illusion of stability while failing to address the issue of welfare damages associated with famine and institutional collapse that could very well result from the second-order effects of an exchange.

8.4 What Remains

The introduction opened with the story of T.K. Jones and his shovels. One could argue this was the moment American civil defense began to collapse into an absurd self-parody, and the field, in ridiculing Jones, stopped trying to answer the question he had answered badly. While Jones was very misguided about dirt solving our safety concerns, the reflex his absurdity licensed—the assumption that because Jones' answer was ridiculous the question must be too—has held for decades, through the end of civil defense, the atrophy of the literature, the narrowing scope of wargames, and the quiet privatization of preparedness into a bunker market that provides a channel for fear without reducing risk. The question is whether an intervention can reduce the damage of a nuclear war without raising its probability, and this paper asserts that some interventions can, that they are identifiable in advance by characteristics the strategic literature has understood in fragments since the 1960s, and that the best of them cost less than a single

weapon system contract. The resilience literature reached the same destination from the opposite direction: the best shelter is a functional and resilient society.⁹¹

Keeping society functional through the most destructive phases of catastrophic scenarios is not a repudiation of prevention, rather it is what taking prevention's possible failure seriously looks like. It requires no illusion of safety, no vision of survival beyond the bounds of responsible thinking—and, most importantly, no shovels. It requires, as it always has, the willingness to directly consider failure modes of deterrence: the one thing the nuclear field has been the least willing to do time and time again.

⁹¹ Janne M. Korhonen. "Sheltering humanity against x-risk: report from the SHELTER weekend." Effective Altruism Forum. Oct. 2022.

A Technical Appendix

This appendix collects the proofs of the propositions stated in Section 3.5.

Proof of Proposition 3.1. Every damage level in the support of ϕ_D satisfies $D' \geq D$, and P is nonincreasing, so $P(D') \leq P(D)$ throughout the integral. Hence

$$\int_D^{D_{\max}} P(D') \phi_D(D') dD' \leq P(D) \int_D^{D_{\max}} \phi_D(D') dD' = P(D), \quad (37)$$

and multiplying by $N(1 - A) \leq 1$ preserves the inequality. $\square$

Proof of Proposition 3.2. Substituting (21) into (20) and integrating the power,

$$\begin{aligned} \int_D^{D_{\max}} \left(\frac{D'}{D_0} \right)^{1-\alpha} \beta D^\beta D'^{-\beta-1} dD' &= \beta D^\beta D_0^{\alpha-1} \int_D^{D_{\max}} D'^{-\alpha-\beta} dD' \\ &= \frac{\beta}{\alpha + \beta - 1} D_0^{\alpha-1} D^{1-\alpha} \left[1 - \left(\frac{D}{D_{\max}} \right)^{\alpha+\beta-1} \right], \end{aligned} \quad (38)$$

which converges because $\alpha + \beta > 1$. Dividing by the normalization $1 - t^\beta$ and by $P(D) = (D/D_0)^{1-\alpha}$ gives (22). The small- t limit is immediate. For $t \rightarrow 1$ both numerator and denominator vanish; by l'Hôpital the ratio tends to $(\alpha + \beta - 1)/\beta$, so $c \rightarrow 1$. For monotonicity write $t = e^{-x}$ with $x > 0$ and $f(x) = (1 - e^{-ax})/(1 - e^{-bx})$, $a = \alpha + \beta - 1 > b = \beta$; then

$$\frac{d \ln f}{dx} = \frac{a}{e^{ax} - 1} - \frac{b}{e^{bx} - 1}, \quad (39)$$

and the map $y \mapsto y/(e^{xy} - 1)$ is decreasing in y for fixed $x > 0$ (its derivative has numerator $e^{xy}(1 - xy) - 1 < 0$), so $d \ln f / dx < 0$: f decreases in x , i.e. c increases in t and hence in D . Finally, we have

$$\ln T[P] = \ln N(1 - A) + \ln c + (1 - \alpha) \ln(D/D_0)$$

, so differentiating with respect to the $\ln D$ yields

$$\frac{d \ln T[P]}{d \ln D} = (1 - \alpha) + \frac{d \ln c}{d \ln D}.$$

On the right hand side, we see that the first term is the naive slope, $-(\alpha - 1)$, while the second is positive by virtue of c being monotone increasing in D , shown above. The transformed curve is still downward-sloping, with the transformation averaging P over damage levels that shift upward as D grows, meaning that $T[P]$ inherits P 's monotone decrease. Hence, proceed via taking magnitudes:

$$\left| \frac{d \ln T[P]}{d \ln D} \right| = (\alpha - 1) - \frac{d \ln c}{d \ln D} < \alpha - 1,$$

meaning that the magnitude of the slope is indeed strictly reduced. $\square$

AI Assistance Acknowledgment

The authors received assistance Claude Opus 5 and Fable 5 to compose the graphs used throughout Section 3 and the proofs for Propositions 3.1 and 3.2 in Appendix A, to render the mathematical derivations more clear throughout all sections in which they are used, and to cite claims previously made by other authors throughout the paper.

Bibliography

- Averch, Harvey A. *Strategic Ambiguity, Asymmetry and Arms Control: Some Basic Considerations*. RM-3426-PR. Santa Monica, CA: RAND Corporation, March 1963. apps.dtic.mil/sti/pdfs/AD0404010.pdf
- Baker, Tom. “On the Genealogy of Moral Hazard.” *Texas Law Review* 75, no. 2 (1996): 237–292. repository.law.upenn.edu/Documents/Detail/on-the-genealogy-of-moral-hazard/401175.
- Benson, Brett V. *Constructing International Security: Alliances, Deterrence, and Moral Hazard*. New York: Cambridge University Press, 2012.
- Braumoeller, Bear F. *Only the Dead: The Persistence of War in the Modern Age*. New York: Oxford University Press, 2019.
- Brodie, Bernard. “The Absolute Weapon: Atomic Power and World Order.” *Yale Institute of International Studies*. New Haven Connecticut. February 15, 1946. osti.gov/opennet/servlets/purl/16380564-wvLB09/16380564.pdf
- Buddemeier, Brooke R., and Michael B. Dillon. *Key Response Planning Factors for the Aftermath of Nuclear Terrorism*. LLNL-TR-410067. Livermore, CA: Lawrence Livermore National Laboratory, 2009.
- Cederman, Lars-Erik. “Modeling the Size of Wars: From Billiard Balls to Sandpiles.” *American Political Science Review* 97, no. 1 (2003): 135–150.
- “Civil Preparedness Review Part 1: Emergency Preparedness and Industrial Mobilization.” Report (Chairman William Proxmire). *Joint Committee on Defense Production, Congress of the United States*. February 1977. babel.hathitrust.org/cgi/pt?id=umn.31951002826765f&seq=7
- Clauset, Aaron. “On the Frequency and Severity of Interstate Wars.” In *Lewis Fry Richardson: His Intellectual Legacy and Influence in the Social Sciences*, edited by Nils Petter Gleditsch, 113–27. Cham: Springer, 2020. doi.org/10.1007/978-3-030-31589-4_10.
- Clauset, Aaron. “Trends and Fluctuations in the Severity and Frequency of Interstate Wars.” *Science Advances* 4, no. 2 (2018): eaao3580. doi.org/10.1126/sciadv.aao3580.
- Cohen, Amichai, and Mirit Lavi. “Operation Rising Lion: A Special Home Front Situation.” Israel Democracy Institute, March 2, 2026. en.idi.org.il/articles/63607.
- Dembe, Allard, and Leslie Boden. “Moral Hazard: A Question of Morality?” *New Solutions* 10, no. 3 (2000): 257–279. pubmed.ncbi.nlm.nih.gov/17208855/.
- Denkenberger, David C. & Joshua M. Pearce. “Cost-effectiveness of interventions for alternate food in the United States to address agricultural catastrophes.” *International Journal of Disaster Risk Reduction*. Vol. 27, March 2018, Pages 278–289. doi.org/10.1016/j.ijdr.2017.10.014

- Denkenberger, David C., and Joshua M. Pearce. *Feeding Everyone No Matter What: Managing Food Security after Global Catastrophe*. Waltham, MA: Academic Press, 2015.
- Do, T., Kim Anderson, and B. Wade Brorsen. “The World’s Wheat Supply.” Oklahoma Cooperative Extension Service, 2010.
- Ermarth, Fritz W. “Contrasts in American and Soviet Strategic Thought.” *International Security* 3, no. 2 (Fall 1978): 138–155.
- Fabian, Emanuel. “IDF Reiterates Safe Rooms Still Best Option Against Iran’s Missiles, Despite 2 Fatalities.” *Times of Israel*, June 16, 2025.
- Federal Emergency Management Agency. “Fiscal Year 2025 FEMA Standard Terms and Conditions.” Fact sheet. August 4, 2025. fema.gov/fact-sheet/fiscal-year-2025-fema-standard-terms-and-conditions.
- Federal Emergency Management Agency. *Taking Shelter from the Storm: Building or Installing a Safe Room for Your Home*. FEMA P-320, 5th ed. Washington, DC: FEMA, 2021.
- Federal Office for Civil Protection. *Shelters: Their Purpose, Construction and Utilisation*. Bern: FOCP, June 2023.
- García Martínez, Juan B., Jeffray Behr, Joshua Pearce, and David Denkenberger. “Resilient Foods for Preventing Global Famine: A Review of Food Supply Interventions for Global Catastrophic Food Shocks Including Nuclear Winter and Infrastructure Collapse.” *Critical Reviews in Food Science and Nutrition* (2025). doi.org/10.1080/10408398.2024.2431207.
- Garrett, Ralph L. “Civil Defense and the Public: An Overview of Public Attitude Studies.” Research Report No. 17. Washington, DC: Mitigation and Research Directorate, Federal Emergency Management Agency. 1979. babel.hathitrust.org/cgi/pt?id=buf.39072020443768&seq=1
- Ghamari-Tabrizi, Sharon. “The Worlds of Herman Kahn: The Intuitive Science of Thermonuclear War.” Harvard University Press. 2005.
- Glasstone, Samuel, and Philip J. Dolan. *The Effects of Nuclear Weapons*. 3rd ed. Washington, DC: U.S. Department of Defense and U.S. Energy Research and Development Administration, 1977.
- Gouré, Leon. *War Survival in Soviet Strategy: USSR Civil Defense*. Coral Gables, FL: Center for Advanced International Studies, University of Miami, 1976.
- Gross, Terry. “In the Event of Attack, Here’s How the Government Plans ‘To Save Itself.’” *Fresh Air*. NPR, June 21, 2017. [npr.org/2017/06/21/533711528/in-the-event-of-attack-heres-how-the-government-plans-to-save-itself](https://www.npr.org/2017/06/21/533711528/in-the-event-of-attack-heres-how-the-government-plans-to-save-itself).
- Hellman, Martin E. “Probabilistic Risk Assessment.” Chapter 4. Laurel, MD: Johns Hopkins University Applied Physics Laboratory, 2022. jhuapl.edu/sites/default/files/2022-12/Ch4.Hellman.pdf.
- Hinz, Fabian. “Iran’s Shattered Missile Strategy.” *Survival* 67, no. 4 (2025). doi.org/10.1080/00396338.2025.2534280.
- Jägermeyr, Jonas, Alan Robock, Joshua Elliott, Christoph Müller, Lili Xia, et al. “A Regional Nuclear Conflict Would Compromise Global Food Security.” *Proceedings of the National Academy of Sciences* 117, no. 13 (2020): 7071–7081.
- Jehn, Felix U., Łukasz G. Gajewski, Jonas Hedlund, Clemens W. Arnscheidt, Lili Xia, Nico Wunderling, and David Denkenberger. “Food Trade Disruption after Global Catastrophes.” Preprint, EarthArXiv, 2024.

- Kahn, Herman. "Report on a Study of Non-Military Defense." *RAND*. July 1958. rand.org/content/dam/rand/pubs/reports/2008/R322.pdf
- Kahn, Herman. "Thinking About the Unthinkable in the 1980s." 1982.
- Kahn, Herman, Erich Fromm, and Michael Maccoby. "The Question of Civil Defense — A Debate." *Commentary*, 1962.
- Kaiser, Robert G. "Civil Defense in Russia Seen As Ineffective: Extra Civil Defense Outlay Held Waste." *The Washington Post Archives*. May 16, 1977. [washingtonpost.com/archive/politics/1977/05/17/civil-defense-in-russia-seen-as-ineffective/cb319485-13db-422d-8f5f-90242324a37e/](https://www.washingtonpost.com/archive/politics/1977/05/17/civil-defense-in-russia-seen-as-ineffective/cb319485-13db-422d-8f5f-90242324a37e/)
- Karger, Ezra, Josh Rosenberg, Zachary Jacobs, Molly Hickman, Rose Hadshar, Kayla Gamin, Taylor Smith, Bridget Williams, Tegan McCaslin, Stephen Thomas, and Philip E. Tetlock. *Forecasting Existential Risks: Evidence from a Long-Run Forecasting Tournament*. FRI Working Paper #1. Forecasting Research Institute, July 2023. forecastingresearch.org/research/existential-risk-persuasion-tournament.
- Kincade, William H. "Repeating History: The Civil Defense Debate Renewed." *International Security* 2, no. 3 (Winter 1978): 99–120.
- Korda, Matt, Eliana Johns, Mackenzie Knight-Boyle, and Hans Kristensen. "The Aftermath: The Expiration of New START and What It Means For Us All." The Federation of American Scientists (Washington, D.C.). February 2026. fas.org/publication/the-expiration-of-new-start/
- Kristensen, Hans, Matt Korda, Eliana Johns, Mackenzie Knight-Boyle, and Kate Kohn. "Status of World Nuclear Forces." The Federation of American Scientists (Washington, D.C.). March 2025. fas.org/initiative/status-world-nuclear-forces/
- Laio, Francesco, Luca Ridolfi, and Paolo D'Odorico. "The Past and Future of Food Stocks." *Environmental Research Letters* 11, no. 3 (2016): 035010. doi.org/10.1088/1748-9326/11/3/035010.
- Lair, Sam. "Shallow Ramparts: Air and Missile Defenses in the June 2025 Israel-Iran War." Foreign Policy Research Institute, October 17, 2025.
- Leaning, Jennifer, and Langley Keyes, eds. *The Counterfeit Ark: Crisis Relocation for Nuclear War*. Cambridge, MA: Ballinger, 1984.
- Lempert, Robert J., Steven W. Popper, and Steven C. Bankes. Shaping the Next One Hundred Years: New Methods for Quantitative, Long-Term Policy Analysis. *RAND Corporation*. (Report) 2003. rand.org/pubs/monograph_reports/MR1626.html
- Lieber, Keir A., and Daryl G. Press. "The New Era of Counterforce: Technological Change and the Future of Nuclear Deterrence." *International Security* 41, no. 4 (Spring 2017): 9–49.
- Lin-Greenberg, Erik, Reid B. C. Pauly, and Jacquelyn G. Schneider. "Wargaming for International Relations Research." *European Journal of International Relations* 28, no. 1 (2022): 83–109.
- Masco, Joseph. "The Future of Fallout and Other Episodes in Radioactive World Making." Duke University Press. 2021.
- McNamara, Robert. "The No Cities Doctrine." Speech. *The University of Michigan*. July 9, 1962. Accessed through *Atomic Archive*. atomicarchive.com/resources/documents/deterrence/no-cities-speech.html

- Mendoza, Martha. “Nuclear Bunker Sales Increase, despite Expert Warnings They Aren’t Going to Provide Protection.” Associated Press, December 2024. apnews.com/article/nuclear-bunkers-war-atomic-bombs-0356fa5b34067c138c64b9143f73c308.
- Moersdorf, Jessica, Morgan Rivers, David Denkenberger, Lutz Breuer, and Florian Ulrich Jehn. “The Fragile State of Industrial Agriculture: Estimating Crop Yield Reductions in a Global Catastrophic Infrastructure Loss Scenario.” *Global Challenges* 8, no. 1 (2024): 2300206. doi.org/10.1002/gch2.202300206.
- National Academies of Sciences, Engineering, and Medicine. “Potential Environmental Effects of Nuclear War — New Report.” Media advisory, June 25, 2025. nationalacademies.org/news/potential-environmental-effects-of-nuclear-war-new-report.
- Newman, M. E. J. “Power Laws, Pareto Distributions and Zipf’s Law.” *Contemporary Physics* 46, no. 5 (2005): 323–351.
- Pape, Robert A. *Bombing to Win: Air Power and Coercion in War*. Ithaca, NY: Cornell University Press, 1996.
- Pauly, Reid B. C. “Would U.S. Leaders Push the Button? Wargames and the Sources of Nuclear Restraint.” *International Security* 43, no. 2 (Fall 2018): 151–192.
- Pearce, Joshua M. & David C. Denkenberger. “A National Pragmatic Safety Limit for Nuclear Weapon Quantities.” *Safety* 2018, 4(2), 25; <https://doi.org/10.3390/safety4020025>.
- Peek, Lori. “Children and Disasters: Understanding Vulnerability, Developing Capacities, and Promoting Resilience — An Introduction.” *Children, Youth and Environments* 18, no. 1 (2008): 1–29. hazards.colorado.edu/uploads/basicpage/Peek%20et%20al%202018%20Children%20and%20Disasters.pdf
- Phillips-Robins, Alasdair. “Catastrophic Risk, Uncertainty, and Agency Analysis.” Institute for Law and AI. Sept. 2022. law-ai.org/catastrophic-risk-uncertainty-and-agency-analysis/
- Pipes, Richard. “Why the Soviet Union Thinks It Could Fight and Win a Nuclear War.” *Commentary* 64, no. 1 (July 1977): 21–34.
- Posner, Richard A. “Efficient Responses to Catastrophic Risk.” 6. *Chicago Journal of International Law*. 511. 2006. chicagounbound.uchicago.edu/cgi/viewcontent.cgi?article=2895&context=journal_articles
- Quarantelli, E. L. “The Sociology of Panic.” Preliminary Paper #283. Newark, DE: Disaster Research Center, University of Delaware, 2001. <https://udspace.udel.edu/bitstreams/981d657d-b864-42e7-ae35-f0b7a969fe85/download>.
- Ravenal, Earl C. “Counterforce and Alliance: The Ultimate Connection.” *International Security* 6, no. 4 (Spring 1982): 26–43. muse.jhu.edu/article/446729.
- Reuters. “Switzerland Plans Revamp of Cold War-Era Nuclear Bunker Network.” December 12, 2024.
- Richardson, Lewis Fry. “Variation of the Frequency of Fatal Quarrels with Magnitude.” *Journal of the American Statistical Association* 43, no. 244 (1948): 523–546.
- Rodriguez, Luisa. “How many people would be killed as a direct result of a US-Russia nuclear exchange?” *Rethink Priorities*. June 30, 2019. rethinkpriorities.org/research-area/how-many-people-would-be-killed-as-a-direct-result-of-a-us-russia-nuclear-exchange/
- Rosenberg, David Alan. “The Origins of Overkill: Nuclear Weapons and American Strategy, 1945–1960.” *International Security* 7, no. 4 (Spring 1983): 3–71.

- Rosenberger, James. “Measuring Community Significance and State Vulnerability after Nuclear Winter.” Working paper, Existential Risk Laboratory, The University of Chicago, August 2024.
- Ruhl, Christian. “Philanthropy to the Right of Boom.” Founders Pledge, February 2023. founderspledge.com/research/philanthropy-to-the-right-of-boom.
- Rump, A., D. Stricklin, A. Lamkowski, S. Eder, M. Port. “Benefit-Cost Analysis of Radiocesium Decorporation by a Prussian Blue Treatment and Stockpiling.” *Journal of Drug Research*, 68(2), 89–99, 2018. doi.org/10.1055/s-0043-119793.
- Sarkees, Meredith Reid, and Frank Wayman. *Resort to War: 1816–2007*. Washington, DC: CQ Press, 2010. Correlates of War Inter-State War Data v4.0, correlatesofwar.org/datasets/cow-war/.
- Sayler, Kelley M. *Hypersonic Weapons: Background and Issues for Congress*. CRS Report R45811. Washington, DC: Congressional Research Service, updated August 27, 2025. <https://www.congress.gov/crs-product/R45811>.
- Scheer, Robert. “Thinking Tuna Fish, Talking Death: Essays on the Pornography of Power.” *With Enough Shovels* (Essay). 1982. digitalcommons.nyls.edu/cgi/viewcontent.cgi?article=1158&context=journal_of_international_and_comparative_law
- Schwartz, Benjamin E. *Right of Boom: The Aftermath of Nuclear Terrorism*. New York: Overlook Press, 2015.
- Speier, Richard, George Nacouzi, Carrie A. Lee, and Richard M. Moore. *Hypersonic Missile Nonproliferation: Hindering the Spread of a New Class of Weapons*. Santa Monica, CA: RAND Corporation, 2017.
- Tannenwald, Nina. *The Nuclear Taboo: The United States and the Non-Use of Nuclear Weapons since 1945*. Cambridge: Cambridge University Press, 2007.
- U.S. Census Bureau. *American Housing Survey*. census.gov/programs-surveys/ahs.
- U.S. Department of Health and Human Services. *HHS Standard Values for Regulatory Analysis, 2024*. Washington, DC: Office of the Assistant Secretary for Planning and Evaluation, January 2024.
- U.S. Department of Transportation. *Departmental Guidance: Treatment of the Value of Preventing Fatalities and Injuries in Preparing Economic Analyses*. Washington, DC: U.S. Department of Transportation, March 2021.
- U.S. Environmental Protection Agency. “Mortality Risk Valuation.” epa.gov/environmental-economics/mortality-risk-valuation.
- “United States Statutes At Large: Containing the Laws and Concurrent Resolutions Enacted During the Second Session of the Eighty-First Congress.” The Federal Civil Defense Act of 1950. *United States Congress*. Vol 64. 1950-1951. hsdl.org/c/abstract/?docid=456688
- United States Strategic Bombing Survey. *The Effects of Atomic Bombs on Hiroshima and Nagasaki*. Washington, DC: U.S. Government Printing Office, 1946.
- Waselenko, Jamie K., Thomas J. MacVittie, William F. Blakely, et al. “Medical Management of the Acute Radiation Syndrome: Recommendations of the Strategic National Stockpile Radiation Working Group.” *Annals of Internal Medicine* 140, no. 12 (2004): 1037–1051.
- Wittner, Lawrence S. “The Nuclear Freeze and Its Impact.” *Arms Control Today* 40, no. 10 (December 2010). armscontrol.org/act/2010-12/nuclear-freeze-and-its-impact.

- World Health Organization. *Iodine Thyroid Blocking: Guidelines for Use in Planning for and Responding to Radiological and Nuclear Emergencies*. Geneva: World Health Organization, 2017.
- Xia, Lili, Alan Robock, Kim Scherrer, Cheryl S. Harrison, Benjamin Leon Bodirsky, Isabelle Weindl, Jonas Jägermeyr, Charles G. Bardeen, Owen B. Toon, and Ryan Heneghan. “Global Food Insecurity and Famine from Reduced Crop, Marine Fishery and Livestock Production due to Climate Disruption from Nuclear War Soot Injection.” *Nature Food* 3 (2022): 586–596.
- Yoshpe, Harry B. “Our Missing Shield: The U.S. Civil Defense Program in Historical Perspective.” Report Contract No. DCPA 01-79-C-0294. *Federal Emergency Management Agency (FEMA)*. April 1981. apps.dtic.mil/sti/tr/pdf/ADA099634.pdf